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# Procuring Impact

## How Public Procurement can Facilitate Change: a Study on Procurement Processes for Electrified Construction.

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by

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Master of Science Thesis TRITA-ITM-EX 2026:644

KTH Industrial Engineering and Management

Industrial Economics and Management

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# Upphandla effekt

Hur offentlig upphandling kan möjligöra förändring: en studie av upphandlingsprocesser för elektrifierat byggande

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## Abstract

This thesis examines how procurement instrument design in electrification-oriented Swedish construction shapes the balance between risk allocation, actor participation and climate ambition. Sustainable public procurement is widely advocated as a transition mechanism, yet the implementation literature documents persistent difficulties in translating climate ambition into practice, particularly when responsibilities stretch across multi-tiered subcontracting chains. The study addresses this gap through a qualitative multi-case study of four Swedish electrification-oriented projects: Slakthusområdet E02, Trafikverket's national bonus model, Mäster Samuelsgatan and Jakobsbergsgatan, and Östra Fyrislund Etapp 5–6. Empirical material draws on semi-structured interviews with procurer-side and contractor-side actors central to these cases, supplemented by procurement documentation and contextual workshop observations.

The findings show that procurement instruments shape the balance between risk, participation and ambition by determining how uncertainty is distributed across utilisation, financing, compliance and enabling-condition risk. Three main insights emerge. First, risk allocation cannot be read from instrument type alone. Similar instruments, such as bonus models or mandatory requirements, distribute risk differently depending on compensation model, dialogue, verification and enabling-condition clarity. The thesis introduces instrument architecture as the analytical unit capturing this configuration. Second, competitive dialogue functions as a temporary institutional learning mechanism that procurers may phase out once internal capability matures. Third, verification burden is not fixed but varies with instrument architecture; binary electric/non-electric requirements are relatively easy to verify, while measured energy use or electrification share-based contracts require substantially more data infrastructure. The findings also indicate that whether electrification ambition is institutionally anchored or carried by individual initiative shapes the durability of demand across procurements.

The thesis contributes a structured framework for reading procurement instrument choice as a risk-allocation decision and offers practical implications for procurers operating in early-stage technology contexts. Limitations include the four-case Swedish scope, the absence of voices from the extended value chain, the rapid market maturation across the cases, and the interpretative rather than measured nature of the analysis.


**Keywords:** public procurement, electrification, construction, risk sharing, instrument architecture, sustainable public procurement, competitive dialogue

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## Sammanfattning

Denna uppsats undersöker hur utformningen av upphandlingsinstrument i elektrifieringsorienterat svenskt anläggningsbyggande påverkar balansen mellan riskallokering, aktörsdeltagande och klimatambition. Hållbar offentlig upphandling lyfts ofta fram som en omställningsmekanism, men implementeringslitteraturen visar på återkommande svårigheter med att omsätta klimatambitioner i praktiken, särskilt när ansvar sträcker sig över flernivåbaserade underentreprenörskedjor. Studien adresserar detta gap genom en kvalitativ flerfallsstudie av fyra svenska elektrifieringsorienterade projekt: Slakthusområdet E02, Trafikverkets nationella bonusmodell, Mäster Samuelsgatan och Jakobsbergsgatan, samt Östra Fyrislund Etapp 5–6. Det empiriska materialet bygger på semistrukturerade intervjuer med centrala aktörer på beställar- och entreprenörssidan, kompletterat med upphandlingsdokumentation och workshopobservationer.

Resultaten visar att upphandlingsinstrument formar balansen mellan risk, deltagande och ambition genom att avgöra hur osäkerhet fördelas mellan nyttjanderisk, finansieringsrisk, efterlevnadsrisk och risk kopplad till möjliggörande villkor. Tre huvudsakliga insikter framkommer. För det första kan riskallokering inte förstås utifrån instrumenttyp enbart. Liknande instrument, såsom bonusmodeller eller obligatoriska krav, fördelar risk på olika sätt beroende på ersättningsmodell, dialog, verifiering och tydligheten i de möjliggörande villkoren. Uppsatsen introducerar instrumentarkitektur som den analytiska enhet som fångar denna konfiguration. För det andra fungerar konkurrenspräglad dialog som en tillfällig mekanism för institutionellt lärande som upphandlande aktörer kan fasa ut när den interna förmågan mognar. För det tredje är verifieringsbördan inte fast, utan varierar med instrumentarkitekturen; binära krav på eldrift respektive icke-eldrift är relativt enkla att verifiera, medan kontrakt baserade på uppmätt energianvändning eller elektrifieringsandel kräver betydligt mer datainfrastruktur. Resultaten visar även att huruvida elektrifieringsambitionen är institutionellt förankrad eller bärs av individuellt initiativ påverkar efterfrågans varaktighet över flera upphandlingar.

Uppsatsen bidrar med ett strukturerat ramverk för att förstå val av upphandlingsinstrument som ett beslut om riskallokering och erbjuder praktiska implikationer för upphandlande aktörer som verkar i tidiga teknologiska utvecklingsskeden. Begränsningar innefattar studiens svenska fyrfallskontext, avsaknaden av röster från den utvidgade värdekedjan, den snabba marknadsmognaden mellan fallen och analysens tolkande snarare än mätbaserade karaktär.


**Nyckelord:** offentlig upphandling, elektrifiering, byggsektorn, riskfördelning, instrumentarkitektur, hållbar upphandling, konkurrenspräglad dialog

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# List of abbreviations

LOU – Lagen om offentlig upphandling (Swedish Public Procurement Act)

SPP – Sustainable Public Procurement

GPP – Green Public Procurement

KPD – Konkurrenspräglad Dialog (Competitive Dialogue)

OEM – Original Equipment Manufacturer

CAPEX – Capital Expenditure

CO2e – Carbon dioxide equivalent

HVO – Hydrotreated Vegetable Oil

# Foreword

This thesis is the result of work carried out during the spring of 2026 as part of the master's programme in Industrial Engineering and Management at KTH Royal Institute of Technology. The research was conducted in connection with the SEEK research project, which provided valuable access to ongoing discussions on electrification in the Swedish construction sector.

I would like to thank my supervisor at KTH, Adam Uhrdin, for his guidance, careful reading, and constructive feedback throughout the research process. I am also grateful to the interviewees from procuring organisations and contractors who generously shared their time, experiences, and reflections. Their openness made the empirical core of this thesis possible. Thanks are also due to those involved in the SEEK project for sharing their insights and to fellow students whose comments at seminars helped sharpen the analysis.

Stockholm, May 2026

Samuel Dagermark

# 1. Introduction

Transport services and construction machinery contribute substantially to greenhouse-gas emissions and account for roughly one fifth of Sweden's climate impact (*Elektrifiering av byggarbetsplatsen*, 2025). Although the technology for electric trucks and machines is increasingly verified and ready, the transition towards fully emission-free construction projects has progressed slowly with electric constituting just 1.5 % of sales in the EU (O'Sullivan, 2025). This is not only due to technical barriers, but additionally because procurement models and the underlying business logic are poorly aligned with the high upfront investments required (O'Sullivan, 2025). This is corroborated by the Energimyndigheten (2025) who claim that trucks for the construction sector can be electrified technically but highlights the issue with distribution of responsibility, where a lot of the climate investments fall to subcontractors, often individual drivers owning their own truck. Therefore it is important for the investment responsibility to move higher up the value chain if the transition is to be successful. In parallel to this thesis the project SEEK (Snabbspår för Elektrifierade Entreprenader genom Klimatupphandling; Fast-Track for Electrified Contracts through Climate Procurement), is being conducted with the overarching ambition to develop procurement and investment models for Stockholm that are economically sustainable, attractive to both public and private clients, and importantly non-excluding for smaller actors. Although this thesis is not directly linked to that project it has guided the direction of the research and will provide a setting for applying the findings. As part of the SEEK project, workshops with stakeholders relevant to the subject matter were attended to improve the understanding of public procurement development for electrification.

Public procurement is one of the main economic activities of governments and refers to the acquisitions of goods and services by governments or public sector entities (Witjes and Lozano, 2016). The general process involves offering a contract to private actors who deliver the product or service for the public entity, and it is a form of bidding process where private actors can compete on price in order to meet pre-set demands from the public actor (*The Public Procurement Act in brief*, 2017). The procurer needs to award the contract to the tender that is economically most advantageous in accordance with EU Public Procurement Directive (Witjes and Lozano, 2016). These pre-set demands tend to be of a technical nature, what result needs to be delivered, but recently additions ascertaining to sustainability have been included as additional criteria. This inclusion forms what is known as Sustainable Public Procurement (SPP) (Witjes and Lozano, 2016). Given the scale and importance of public procurement, it has the potential to greatly increase sustainability both through its immediate impact in each procurement and through influencing the private sector into becoming more sustainable (Brammer and Walker, 2011). As such it is important to shed light on how this can be used as a tool for transitions in the context of construction.

Instruments such as requirements, bonuses, penalty schemes and follow-up routines can, in practice, shift financial and performance risk across the supply chain, thereby influencing who can participate and invest. Smaller actors (e.g., small hauliers and machine owners) may face binding constraints related to liquidity, financing, utilisation uncertainty, administrative compliance burdens, and exposure to penalties if enabling conditions are uncertain.

## 1.1 Literature overview

A large share of empirical work in Green Public Procurement (GPP) and Sustainable Public Procurement (SPP) relies on surveys and qualitative methods rather than quantitative outcome evaluation, which shapes what kinds of claims the field typically makes (e.g., perceived barriers, organisational practices) and limits causal inference about impact. The European Union also appears as a dominant empirical setting, often explained by Europe's position as an early mover in GPP and the scale and institutionalisation of procurement within the EU context (Cheng *et al.*, 2018). Within this body of work, scholars have repeatedly investigated how environmental criteria are integrated into procurement and which organisational and operational barriers limit implementation. However, supplier-side perspectives have often been studied in sectors such as cleaning services, healthcare, and manufacturing-related SMEs (Oruezabala and Rico, 2012; Rizzi *et al.*, 2014; Lundberg *et al.*, 2015). While these studies provide useful insight into suppliers' responses to green procurement demands, the extent to which their findings transfer to construction is uncertain due to the sector's project-based production, fragmented supply chains, and reliance on subcontracting.

Within Swedish construction procurement, existing research frequently points to practical challenges in operationalising "green" ambitions. In particular, studies emphasise issues of measurability and the limited effectiveness of green initiatives when follow-up and evaluation are weak or inconsistent (Varnäs, Balfors and Faith-Ell, 2009). At the same time, there appears to be relatively limited research that centres the experiences and interpretations of contractors and suppliers who must translate procurement requirements into practice. An exception is Kadefors *et al.* (2021), who examine environmental infrastructure procurement and incorporate contractors' views on the Swedish Transport Administration's Klimatkalkyl. Their findings suggest that strict requirements can be excluding for some actors, potentially reducing competition, whereas award criteria may create less exclusionary dynamics (Kadefors *et al.*, 2021). They also highlight that sustainability-related calculations and reporting can increase tendering costs, consuming scarce engineering and specialist time and potentially lowering contractors' interest when alternative projects exist (Kadefors *et al.*, 2021). Together, this points to the importance of examining not only what procurement documents prescribe, but how procurement instruments are interpreted, resourced, and experienced by market actors.

A further complication is that construction differs from traditional manufacturing in ways that shape both risk exposure and the feasibility of innovation adoption. Construction projects involve temporary organisations, on-site production, and one-of-a-kind outputs, which complicate productivity gains and standardisation (Segerstedt and Olofsson, 2010). Main contractors may execute only a limited share of production while suppliers and subcontractors perform substantial portions of work, increasing dependence on subcontractors for flexibility and capacity (Segerstedt and Olofsson, 2010). This structure creates conditions where procurement-driven innovation requirements may not translate straightforwardly through multiple contractual layers, particularly when subcontractor selection is strongly cost-driven (Hartmann and Caerteling, 2010). Research further suggests

that fragmentation and weak cross-firm coordination can hinder innovation and renewal, since innovation often requires combining resources and learning across firm boundaries (Havenvid, Håkansson and Linné, 2016). Against this backdrop, electrification-focused procurement raises a particularly acute adoption challenge: electrified machinery and vehicles often require high up-front investment and expose contractors to financing and utilisation risk (Östgren, Forsgren and Tschiesner, 2019; Yi *et al.*, 2026). The interaction between these electrification-specific investment barriers and procurement instrument design (e.g., penalties, bonuses, and follow-up regimes) remains underexplored, especially from a comparative, case-based perspective that contrasts procurers' intentions with contractors' perceived impacts.

## 1.2 Research gap and purpose

The SPP/GPP literature is extensive and documents common implementation barriers, including administrative complexity, limited follow-up, and challenges in translating environmental ambition into measurable and enforceable practice (Varnäs, Balfors and Faith-Ell, 2009; Cheng *et al.*, 2018). Research on green procurement in construction further highlights difficulties related to measurability and evaluation and indicates that procurement design choices may have unintended consequences for competition and supplier participation (Kadefors *et al.*, 2021). At the same time, the project-based and fragmented nature of construction supply chains means that requirements imposed at the client–main contractor interface may be difficult to transmit through multiple subcontracting layers, particularly when cost competition dominates (Hartmann and Caerteling, 2010; Segerstedt and Olofsson, 2010). This gap is particularly relevant because incentive/penalty schemes and follow-up regimes are central governance mechanisms in contracting. In performance-based contracting, payments (or deductions) are tied to predefined KPIs, which can shift performance and compliance risk toward suppliers and increases the need for credible monitoring and enforcement (Molitoris, Glas and Eßig, 2025). In construction contracting, incentive/disincentive arrangements and risk allocation affect contractor pricing, behaviour, and transaction costs, with misallocated risk often returning as higher contingencies or disputes (Li, Arditi and Wang, 2015). Moreover, risk-shifting dynamics in multi-tier contracting chains can leave lower-tier actors bearing risks they are poorly equipped to manage (McDermott and Hayes, 2018). Finally, public procurement's limited ability to guarantee contract continuity can weaken suppliers' ability to recover innovation investments across projects (Lundberg, Marklund and Strömbäck, 2016). Taken together, this literature examines procurement mechanisms, risk allocation and electrification barriers largely separately. What remains underexplored is how the design of procurement instruments brings these together: how a given design positions electrification-related risk, and how procurers and contractors interpret its implications for participation and feasibility, examined comparatively across cases. The literature does not yet offer an integrated analytical lens connecting instrument design to risk positioning and implementation feasibility in this setting.

**The purpose of this thesis** is to examine how procurement instrument design in electrification-oriented Swedish construction procurements structures risk allocation and responsibility positioning. The study further analyses how procurers and contractors interpret the implications of these designs for participation conditions and implementation feasibility, and where tensions emerge between climate ambition and feasible implementation.

To address this, the thesis develops a conceptual framework (Chapter 3) that takes procurement instrument design as its analytical point of departure and links it to risk positioning, participation conditions and implementation feasibility. The framework is then applied to the four cases and refined in light of the empirical material (Chapter 6).

To examine this, the study draws on four cases: the Slakthusområdet E02 earthworks project in Stockholm (Exploateringskontoret), Trafikverket's national bonus system for electrified construction machinery, the Mäster Samuelsgatan and Jakobsbergsgatan street reconstruction in Stockholm (Trafikkontoret), and the Östra Fyrislund Etapp 5-6 development project in Uppsala (Uppsala Kommun). These cases were selected to represent variation in instrument design, covering requirement-based, bonus-based, penalty-heavy, and combined approaches across both municipal and national procurement contexts. Empirical data consists primarily of interviews with procurers and main contractors directly involved in each procurement, supplemented by procurement documents.

## 1.3 Research questions

**Main RQ***:* How does the design of procurement instruments in electrification-oriented Swedish construction shape the balance between risk allocation, actor participation, and climate ambition?

**Sub-Q1***:* What procurement instrument designs are employed in the selected cases, and how do these designs position electrification-related risks and responsibilities?

**Sub-Q2:** How do procurers and contractors interpret the implications of different procurement instrument designs for participation conditions and implementation feasibility?

**Sub-Q3:** What tensions emerge between climate ambition and implementation feasibility across the selected cases?

## 1.4 Scientific contribution

This study contributes to the emerging literature on climate procurement as a transition mechanism by analysing *how* procurement instrument design structures risk allocation and shapes participation conditions and implementation feasibility. Empirically, it adds comparative Swedish multi-case evidence spanning municipal and national infrastructure procurement contexts. Analytically, it develops a structured framework linking instrument design to risk positioning and feasibility outcomes, examined from both procurer and contractor perspectives. The contribution is intended to support both academic understanding and the SEEK project's procurement model development.

## 1.5 Delimitations

The thesis is delimited to four Swedish construction procurements that incorporate electrification-related instruments, spanning municipal and national infrastructure procurement contexts. The focus is on procurement instrument design, risk allocation, and feasibility rather than on the technical performance of electric vehicles or machines. The thesis will not quantitatively model total cost of ownership or emissions outcomes; instead, it examines participation conditions and implementation feasibility through qualitative evidence from interviews with procurers and main contractors, supplemented by procurement documents. The study is further delimited in terms of actor coverage: it does not claim to directly represent the perspectives of subcontractors, machine owners, or other lower-tier supply chain actors, though the analytical framework remains attentive to how procurement designs may affect such actors.

# 2. Literature Review

This chapter reviews literature on public procurement, sustainable public procurement, construction electrification, contracting and risk allocation. Together, these strands form the basis for the conceptual framework developed in Chapter 3.

## 2.1 Public Procurement in Sweden

In Sweden, public procurement is governed by the Swedish Public Procurement Act (LOU) and is intended to make efficient and legally certain use of public funds through competition on equal terms (*Public procurement – How it works in Sweden*, 2015; *The Public Procurement Act in brief*, 2017). It is also expected to promote innovation and to take environmental and social concerns into account. EU directives extend equal treatment across borders and prohibit favouring national suppliers (Brammer and Walker, 2011). Public procurement differs from private purchasing in that the process is documented and publicly accessible, and in that a procuring organisation cannot continue with a supplier on subsequent projects without a new procurement, which ensures equal treatment but limits the continuity available to suppliers (*About Public Procurement*, no date).

## 2.2 Sustainable Public Procurement

Sustainable Public Procurement refers to the integration of broader social and environmental impacts within procurement processes from the public sector (Brammer and Walker, 2011). Similarly Green Public Procurement is the process of public actors procuring goods and services with specifically lower environmental impact than alternatives with similar functionality and has been a part of public purchasing for several decades. It has been included as a part of the EU public procurement directive since 2004 (Sparrevik *et al.*, 2018). Due to the large amounts of money spent by the public sector on construction, through public procurement, the effects of integrating sustainability in these processes can yield big returns (Sourani and Sohail, 2011).

Green Public Procurement is the extension of public procurement where policy driven efforts for environmental sustainability are included as contractual requirements (Witjes and Lozano, 2016; Sparrevik *et al.*, 2018). Although the green public procurement needs to comply with existing procurement law which limits its practical effects as a policy instrument (Sparrevik *et al.*, 2018). Further Sparrevik *et al.* (2018) reflects that the lack of financial obligations in the policy process makes GPP vulnerable to budget constraints at public actors thus limiting their impact. Without strong financial support from the financing actor and cooperation with the contractor these environmental requirements are seen as more of a burden than a possibility for competitive advantage and source for innovation (Sparrevik *et al.*, 2018). The problem with GPP in many cases is the difficulty in achieving cost efficiency, stemming from the mandatory inclusions of environmental criteria (Cheng *et al.*, 2018).

Sparrevik *et al.* (2018) argues that there is a need to explicitly include policy requirements in the procurement process. Although they conclude that there is difficulty in incorporating environmental benefits in decision methods, they discuss a specific case where the internal motivation was to create a "lighthouse" example, showing the way to others. As such they managed to keep the different parties on track despite the lack of official GPP structures in the procurement process.

Some barriers for proper integration of sustainability measures in public procurement processes include lack of funding, restrictions on expenditure, reluctance of incurring higher capital costs; lack of awareness, understanding and information commitment and demand; insufficient policy, regulation and incentives by leadership; lack of sufficient time to address sustainability issues; lack of long term perspective; the perception of sustainability measures incurring greater costs; resistance to change; insufficient integration and link-ups with industry actors; and insufficient research and development (Sourani and Sohail, 2011). Some of these are mirrored by findings by Brammer and Walker (2011) stating for instance that financial constraints are some of the most common barriers; with the sustainably produced products being perceived as more costly to procure. Another barrier is the lack of information for the individual procurement practitioners who without the knowledge of available sustainable alternatives and the lack of clear support from senior management are reluctant to independently introduce sustainability criteria in the procurement process, especially when they might incur increased costs (Brammer and Walker, 2011). This point is supported by Cheng *et al.* (2018) who conclude that the environmental criteria often play second fiddle to quality and cost metrics in the selection process of a procurement process. They state that this might stem from the complication caused by integration of environmental criteria into the selection when the procuring party, often a public authority, lacks the requisite knowledge and competence on the environmental aspects. Further the issue of lax follow-up routines on environmental performance once the contract has been granted introduces the issue of key actors being left without the necessary information (Cheng *et al.*, 2018).

Lundberg, Marklund and Strömbäck (2016) argues that GPP can work as a signalling system (showcasing long-term commitment to change, influencing the supply-side) in the case that the suppliers can distribute needed adjustments and investment costs over a longer time horizon, i.e. stepwise over several procurements. One problem with this process is the slower

adoption this would entail with consequently lower sustainability impacts at first. Another issue with the stepwise change is that in accordance with procurement legislation one supplier cannot be guaranteed subsequent contracts and as such bear the risk stemming from the sustainability investment.

In the supplier selection process, it is important to find relevant suppliers that share the values of the procuring organisation as the cooperation then can move beyond formal contracts and the intended results are more likely to be achieved. Due to the nature of public procurement this is usually achieved through a competitive dialogue, a procedure that allows the public authority to build a relationships with suppliers before the bidding phase through communication with selected contractors (Sönnichsen and Clement, 2020).

Within Swedish construction procurement, research has pointed to persistent challenges around measurability and the real effectiveness of environmental initiatives when they are not paired with robust follow-up and evaluation. A recurring theme is that sustainability ambitions risk remaining symbolic if monitoring, verification, and learning loops are weak, inconsistent, or absent (Varnäs, Balfors and Faith-Ell, 2009).

## 2.3 Electrification adoption in construction

Electric construction machinery is the main facilitator for more environmentally sustainable construction sites, it includes excavators, trucks and other large stationary machinery such as the sorting plant used in Slakthusområdet E02 (*Minst 50 procent eldrift. Ökade krav i entreprenad E02 i Slakthusområdet*, 2024). There are however some barriers to the effective implementation of these electrified substitutes, namely their higher costs and energy supplies. Further there can be issues with knowhow on the construction plant as electric alternatives introduces new requirements and ways of working. Then there is the issue of infrastructure support needed, specifically relating to charging where increased costs can occur. Additionally not all construction sites are suitable for this transition since access to high-voltage power is a prerequisite (Stokke *et al.*, 2023). The subject of higher costs is corroborated by Yi *et al*. (2026), mainly the purchasing cost of the electric construction machinery which are 1.7 to 2.4 times the price higher than traditional equivalents. The barrier of charging time and range can also negatively affect productivity with electric machines (Yi *et al.*, 2024).

Construction differs from traditional manufacturing due to features such as temporary organisations, on-site production, and one-of-a-kind outputs, which complicate standardisation and productivity improvements. In addition, main contractors may perform only a limited share of production themselves, while suppliers and subcontractors execute substantial parts of the work, creating dependency on multi-tier supply chains (Segerstedt and Olofsson, 2010).

Fragmentation and cost-driven subcontractor selection can hinder innovation and renewal in construction by separating industry actors and weakening coordination across firm boundaries. One line of argument is that innovation depends on combining resources and activities across firms; something made difficult in fractured, project-based settings. From

this perspective, developing longer-term relationships, and establishing formal or informal contractual arrangements that support coordination and joint learning, becomes important for changing entrenched practices (Havenvid, Håkansson and Linné, 2016).

## 2.4 Contracting and risk allocation

One contracting alternative that has gained prevalence in recent years is Performance-based contracting (PBC) in which incentives and penalties are used to align suppliers and customers. The base in this contracting process is pre-determined key performance indicator based on what is expected in the transaction. This method ties the payments to the achieved performance and shifts the risk from the customer to the supplier as the payment is contingent on the result being delivered to the supplier in accordance with their expectations (Molitoris, Glas and Eßig, 2025). This could be used for predetermined expectations on sustainability, such as electric machine hours at a construction site in order to incentivise the proper adherence to the contract.

When connecting PBC to public procurement some issues arise as is the case in a case on German procurement as described by Molitoris, Glas, and Eßig (2025) where they identify an issue pertaining to public procurement regulations. In normal PBC situations the supplier is incentivised to innovate and improve their delivery as it can result in bonuses, meanwhile public procurement introduces the issue of least cost operations where the budgetary constraints forbid the inclusion of bonuses as they are perceived as unnecessary even when performance is above expectations. Further one of the main strengths of PBC is the option to negotiate further contracts based on previous performance and as such the innovation and improvement of processes becomes an apt investment for the supplier. In the case of European public procurement this is nullified as the laws intend to increase competition and a contract cannot be extended to the same supplier without a new procurement procedure open to several actors (Molitoris, Glas and Eßig, 2025). Similar conclusions are made by Li, Arditi and Wang (2015) stating that incentives and disincentives are important in construction contracts, ensuring that contractors deliver on time and in accordance with the budget. This is beneficial for both the owner and contractor as they are forced to collaborate and benefit from reduced time and cost spent on the project.

The concept of transaction costs is also relevant to the subject since it affects how costs are split among actors in a value-chain, something critical to how costs are carried in construction projects. These transaction costs can be split into pre-contract and post-contract transaction costs. The pre-contract transaction costs relate to drafting and negotiating a contract, this can include costs of environmental impact assessments, feasibility studies, preliminary design and tender preparation. Post-contract transaction costs include contracts administrative costs, the cost of claims and change orders, incentive payments and costs relating to dispute resolution (Li, Arditi and Wang, 2015).

Fair risk allocation relates to whether the owner or the contractor is carrying risk in a project; some risk can be retained by the owner while some is transferred to the contractor. This typically leads to the contractor raising their markup or contingency which in the end reverts

the price increase back to the owner. Although appropriately distributing risk in accordance with which party is best able to carry said risk can lead to the integrity and acceptance of the contract being improved in addition to the relationship being positively affected. Ultimately fair distribution of risk can reduce the transaction costs by diminishing the prevalence of disputes and conflicts (Li, Arditi and Wang, 2015).

Subcontracting is prevalent in construction, allowing main contractors to gain flexibility, reduce fixed assets and access specialist capacity (Hartmann and Caerteling, 2010; McDermott and Hayes, 2018). It also moves risk down the supply chain. McDermott and Hayes (2018) note that risk-shifting through multiple tiers can leave lower-tier actors carrying risks they are poorly equipped to manage, a dynamic amplified in highly competitive, cost-driven contracting environments where pressure from higher tiers can force disadvantageous risk-bearing. Subcontractor selection itself often remains strongly price-driven, which can admit new entrants but also concentrate compliance and cost risk on actors with limited capacity to absorb it (Hartmann and Caerteling, 2010; Molitoris, Glas and Eßig, 2025). This matters here because procurement requirements imposed at the client and main-contractor interface may not transmit cleanly through these layers.

One contractor-facing insight in the Swedish context is that the use of strict requirements can be excluding for certain actors and may reduce competition, whereas award criteria can be less exclusionary because they allow contractors to propose differing solutions rather than being screened out at the outset. Another contractor-facing issue concerns the additional tendering costs created by sustainability calculations and reporting requirements; these can consume limited engineering and carbon-specialist time and potentially reduce contractors' interest in bidding when other projects are available (Kadefors *et al.*, 2021).

# 3. Conceptual Framework

As identified in Chapter 1, existing research tends to examine procurement mechanisms, risk allocation and electrification barriers in isolation rather than as an integrated design. This chapter develops a framework to address that gap. The framework is treated as a provisional analytical lens, built from the literature reviewed in Chapter 2 and examined against the empirical material in Chapter 6.

This thesis conceptualises climate-oriented procurement as a mechanism through which public actors may shape the conditions for electrification in construction. More specifically, it examines how procurement design structures incentives, obligations, and responsibilities in ways that affect the feasibility and participation conditions of electrified construction practices.

The framework positions procurement instrument design as the primary analytical point of departure. Different procurement instruments structure contractual expectations in different ways, for example through mandatory requirements, financial incentives, sanctions, or combinations of these. These instruments do not shape electrification directly. Rather, they do so by influencing how electrification-related risks and responsibilities are positioned across

the supply chain, and by shaping how feasible participation appears under different procurement designs.

Risk allocation is therefore treated as the central analytical mechanism linking procurement design to its likely effects. The implications of procurement design are not uniform across actors. Instead, they are conditioned by differences in actor size, organisational capacity, and position in the supply chain, since firms differ in their ability to absorb uncertainty, mobilise capital, and manage operational exposure.

The outcomes of interest are therefore participation conditions and implementation feasibility rather than investment behaviour as such. These concern what participation conditions different procurement designs offer to relevant actors, how feasible electrification appears under different instrument designs, and where tensions arise between climate ambition and market participation conditions.

The analysis draws on three interconnected strands of literature. First, sustainable and green public procurement literature explains how environmental objectives are translated into procurement criteria and identifies barriers that often limit implementation (Varnäs, Balfors and Faith-Ell, 2009; Brammer and Walker, 2011; Cheng *et al.*, 2018). Second, literature on contracting and risk allocation helps explain how contractual design distributes financial, operational, and compliance-related exposure between parties (Li, Arditi and Wang, 2015; McDermott and Hayes, 2018; Molitoris, Glas and Eßig, 2025). Third, literature on construction electrification highlights the sector-specific barriers that make procurement design especially consequential, including high upfront costs, utilisation uncertainty, and dependence on enabling infrastructure (Stokke *et al.*, 2023; Yi *et al.*, 2026). Taken together, these literatures support an analytical framework in which procurement design matters because it structures risk, responsibility, and feasibility in different ways.

## 3.1 Procurement instrument design

In this thesis, procurement instruments are understood as the mechanisms through which procuring entities attempt to influence contractor and supplier behaviour in favour of electrification. Four main instrument types are distinguished: requirements, bonuses, penalties, and combinations.

Requirements establish mandatory thresholds or conditions that bidders must meet either to participate in the tender or to perform the contract. In the context of electrification, such requirements may concern the use of electric machinery, restrictions on fossil fuel use, or broader emission-related standards. Requirements can create a clear performance floor, but they may also raise the threshold for participation where technical capacity or infrastructure is uncertain (Kadefors *et al.*, 2021).

Bonuses provide additional compensation for electrified performance. Unlike requirements, they do not exclude actors who lack immediate electrification capacity, but instead create an economic incentive for voluntary uptake. Bonuses may therefore be particularly relevant in market contexts where technology, charging infrastructure, or supplier maturity remains uncertain (Molitoris, Glas and Eßig, 2025).

Penalties are sanctions attached to non-compliance. Their analytical significance lies in their capacity to increase the consequences of failing to meet electrification-related obligations. Penalties can strengthen enforceability, but they may also intensify perceived risk if compliance depends on factors that contractors only partly control.

Combinations refer to procurement designs that layer several instrument types within the same procurement. Such combinations may attempt to address more than one implementation challenge at once by combining minimum obligations, incentives, and enforcement mechanisms. Their effectiveness depends not simply on the presence of several tools, but on how coherently they work together.

In addition to these four instrument types, the framework recognises two cross-cutting procedural mechanisms that affect how procurement designs operate in practice. Competitive dialogue is treated as a pre-contract mechanism that may reduce information asymmetries, clarify feasibility, and surface implementation constraints before contract award (Sönnichsen and Clement, 2020). Follow-up and verification are treated as post-contract mechanisms that shape how credible and enforceable procurement instruments become during project execution. These two mechanisms are therefore not treated as separate outcomes, but as conditions that influence how procurement design translates into practical risk exposure and implementation feasibility.

## 3.2 Risk allocation and responsibility positioning

The core analytical proposition of this thesis is that procurement instruments matter because they structure how electrification-related risks and responsibilities are positioned across the contractual chain. In other words, procurement design helps determine who is expected to carry the consequences of uncertainty associated with electrification. Four categories of risk are treated as particularly relevant in this study.

Utilisation risk refers to uncertainty about whether electrified equipment will be used sufficiently to justify the costs associated with acquisition or deployment. This is important in a context where electrification often involves substantial upfront costs and where returns depend on sustained use over time (Yi *et al.*, 2026).

Financing and CAPEX risk concerns the ability of firms to access and justify capital for electrified machines, vehicles, or related investments. This risk becomes particularly important where electrification-related investment must be made before demand is stable or where cost recovery depends on future project opportunities (Stokke *et al.*, 2023; Yi *et al.*, 2026).

Compliance and penalty risk arises when contractual obligations are backed by sanctions. This category captures the exposure actors face when non-compliance has financial or contractual consequences, particularly where compliance depends on technical, logistical, or supply-chain conditions that are not fully within their control.

Enabling-condition risk refers to uncertainty regarding the practical prerequisites for electrification, such as charging infrastructure, grid capacity, power supply, or other site-

specific conditions required for electric operation. Since these prerequisites are not always controlled by the same actor that carries the contractual obligation, they are analytically central to understanding how feasible electrification becomes in practice.

Rather than assuming that these risks can be fully traced empirically to every actor involved, this thesis uses the framework to analyse how different procurement designs appear to position such risks within the supply chain. This makes it possible to examine how procurement design may widen or narrow the practical conditions for participation, even where direct access to all actor categories is limited (McDermott and Hayes, 2018; Molitoris, Glas and Eßig, 2025).

## 3.3 Actor size and supply-chain position

Actor size and supply-chain position are retained as an important analytical dimension, treated as a sensitising concept rather than as a fully testable moderating variable across all cases. The rationale is that procurement design is unlikely to have uniform implications for actors with different levels of financial resilience, organisational capacity, and contractual influence.

Larger organisations may be better positioned to absorb uncertainty through diversification, internal resource allocation, and stronger access to finance. Smaller firms, individual machine owners, and lower-tier subcontractors are more likely to experience the same risk exposures as more constraining, since they typically have fewer opportunities to spread costs or hedge uncertainty across projects (Lundberg, Marklund and Strömbäck, 2016; Kadefors *et al.*, 2021).

At the same time, the empirical material available to this study does not provide equal access to all such actor categories. This study therefore does not make exhaustive claims about how all firm types actually behave. Instead, the analysis remains attentive to the possibility that procurement designs may have uneven consequences depending on which actors are expected to absorb the risks and responsibilities associated with electrification. In this way, actor size remains conceptually important without becoming an empirical claim that the study cannot fully support.

## 3.4 Participation conditions and implementation feasibility

The main outcomes of interest in this thesis are participation conditions and implementation feasibility. Participation conditions refer to how accessible a procurement appears to actors operating under different capacities and constraints. This includes whether procurement design appears to widen or narrow the pool of actors able or willing to participate, either directly through eligibility thresholds or indirectly through the level of uncertainty and exposure created.

Implementation feasibility refers to how practically workable electrification appears under a given procurement design. This includes whether relevant responsibilities are clearly placed, whether enabling conditions appear manageable, and whether the balance between ambition and enforceability appears realistic.

These outcomes are shaped by the way actors are likely to interpret the overall balance between expected opportunities and expected exposure. A procurement design that offers strong climate ambition but leaves major uncertainties unresolved may produce different conditions from one that reduces uncertainty through dialogue, support, or clearer responsibility allocation. Similarly, some instruments may primarily affect participation conditions at the bidding stage, while others may more strongly affect the practical feasibility of implementing electrification during contract execution. Non-financial considerations, such as strategic positioning or reputational value, may also shape how actors interpret procurement opportunities, though these are treated as secondary within the broader question of how procurement design structures perceived feasibility, responsibility, and exposure.

## 3.5 Summary of the analytical framework

Taken together, the framework proposes the following analytical logic. Procurement instrument design shapes how climate-related obligations, incentives, and sanctions are structured. These design choices influence how electrification-related risks and responsibilities are positioned across the supply chain. The implications of this positioning are conditioned by actor size and supply-chain position, since different actors have different capacities to absorb uncertainty. The result is variation in participation conditions and implementation feasibility across procurement designs.

Competitive dialogue and follow-up and verification are treated as cross-cutting mechanisms that affect how procurement instruments are interpreted, adjusted, and enforced before and after contract award, respectively.

The framework is used in the analysis to compare how different procurement designs structure feasibility, responsibility, and participation conditions across the selected cases. Rather than claiming to measure the behaviour of all actors directly, it provides a structured lens for examining how procurement design appears to shape the practical conditions under which electrification is pursued. A central analytical move in the thesis is to treat instrument design as more than a question of instrument type. Two procurements may share the same instrument type, for example a bonus model or a mandatory requirement, yet position risk and shape participation very differently depending on the compensation model surrounding it, whether competitive dialogue precedes contract award, how verification is structured, and how clearly the enabling conditions are specified. The thesis refers to this broader configuration as *instrument architecture*, and it is at this level, rather than at the level of instrument type alone, that the analysis is conducted.

Figure 1 summarises the analytical logic used in the thesis, showing how procurement instrument design shapes risk allocation and responsibility positioning, which in turn affects participation conditions and implementation feasibility.

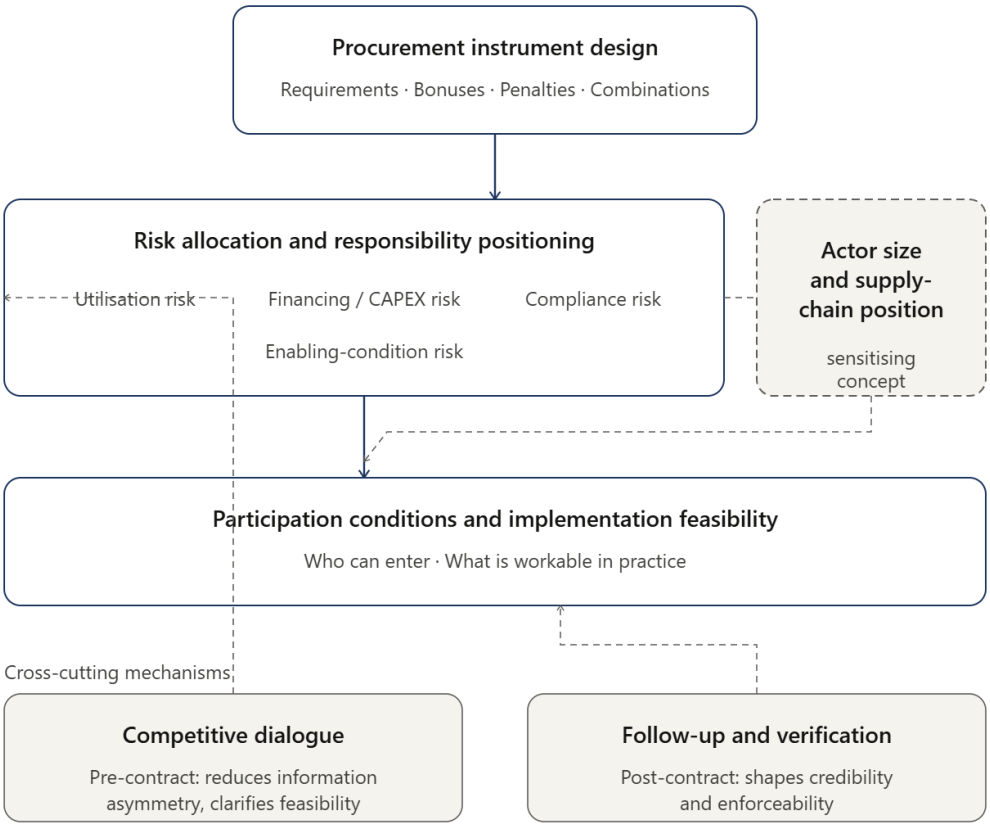



*Figure 1: Conceptual Framework*

# 4. Methodology

This chapter describes the methodology of the thesis, including the research design, case selection, empirical material, data collection and analysis. The study adopts a qualitative multi-case study design. Semi-structured interviews constitute the primary empirical material, while procurement documents are used as a complementary source to analyse the formal design of the procurement instruments. In addition, observations from SEEK workshops are used as contextual material to support the broader understanding of procurement-model development for electrified construction.

The purpose of combining these sources is to examine both how electrification-oriented procurement instruments are formally structured and how they are interpreted by actors involved in or close to their implementation. This is important since the thesis does not only examine what is written in procurement documents, but also how requirements, bonuses, penalties, verification routines and responsibility allocations are understood by procurers and contractors in practice.

## 4.1 Research design

This thesis adopts a qualitative case study design. This approach is suitable due to the complex and context-dependent nature of public procurement for electrified construction. The thesis does not aim to quantitatively measure the climate impact of different procurement instruments or statistically evaluate the effectiveness of requirements, bonuses or penalties. Instead, the purpose is to understand how procurement instrument design structures risk allocation, responsibility positioning, actor participation and implementation feasibility.

Yin (2009) highlights that case study research is suitable when the research questions concern how or why a contemporary phenomenon takes place in a real-life context. This is relevant for the present study, since procurement instruments cannot be fully understood separately from their specific project setting, market maturity, available charging infrastructure, contractual model and actor constellation. In this thesis, the procurement instrument is therefore not treated as an isolated contractual clause, but as part of a wider procurement and implementation context.

Baxter and Jack (2008) describe qualitative case study methodology as a useful approach for studying complex phenomena within their context. They also emphasise the importance of defining the case and using several data sources when appropriate. This corresponds to the design of the present study, where the selected cases are examined through interviews, procurement documents and contextual observations. The case study design is therefore used to create a deeper understanding of how electrification procurement is interpreted and operationalised in practice.

The study is designed as a multi-case study. The purpose of using several cases is not to produce statistical generalisation, but to enable analytical comparison between different procurement designs. All cases concern electrification-oriented construction procurement in Sweden, but they differ in how electrification is governed contractually. This allows the thesis to compare requirement-based, bonus-based, penalty-centred and dialogue-based approaches within a shared public-procurement context.

Before adopting this design, two alternatives were considered. A quantitative or outcome-based evaluation was not pursued, because the effects of these instruments are still emerging and difficult to isolate across heterogeneous projects, and because the research questions concern interpretation and feasibility rather than measurable impact. A single in-depth case was also considered, but several cases were needed to compare how different instrument designs position risk and feasibility.

The study is primarily exploratory, since electrified construction procurement is still an emerging practice. At the same time, it contains explanatory elements, since it seeks to clarify why different procurement instruments may create different implications for risk allocation, participation and feasibility. The analysis therefore moves between empirical observations and the conceptual framework developed in Chapter 3, where procurement instrument design is treated as the analytical starting point.

## 4.2 Case selection

The cases were selected to represent variation in procurement instrument design within the shared context of Swedish electrified construction procurement. Eisenhardt (1989) argues that case selection in case study research can be based on theoretical rather than statistical reasons. This is relevant for the present thesis, since the cases were not selected to be representative of all Swedish construction procurements, but because they provide analytically useful variation in how public actors use procurement to steer electrification.

This follows the diverse-case method (Seawright and Gerring, 2008), in which cases are chosen to span the range of values on the dimension of analytical interest rather than to maximise difference across all conditions. Procurement instrument design can be understood as a configuration of features, combining requirement, bonus, penalty and dialogue elements, and the four cases were selected to represent different points across this range. All cases are situated within the same Swedish public-procurement framework, which provides a common institutional baseline for comparison, while the variation in instrument design allows the study to explore, in line with its exploratory aim, how different designs are associated with different patterns of risk allocation, participation and feasibility.

The four cases included in the thesis are Slakthusområdet E02, Trafikverket's bonus model, Mäster Samuelsgatan and Jakobsbergsgatan, and Östra Fyrislund Etapp 5-6. These cases represent different procurement instrument logics. Slakthusområdet E02 represents a requirement-based approach combined with competitive dialogue. Trafikverket's bonus model represents a bonus-based approach, where electrification is encouraged through financial compensation rather than through a direct electrification requirement. Mäster Samuelsgatan and Jakobsbergsgatan represent a requirement- and penalty-centred approach combined with competitive dialogue and qualitative evaluation. Östra Fyrislund Etapp 5-6 represents a requirement-based approach with a strong penalty structure, where electrification is formulated as a binding contract condition rather than as a voluntary bonus mechanism.

This variation is central to the research questions. The thesis examines how procurement instrument design shapes the balance between risk allocation, actor participation and climate ambition. To answer this, it is necessary to compare cases where the design of the procurement instrument differs. The cases also vary in terms of procuring organisation, project setting, use of market dialogue and follow-up structure. This makes it possible to examine how similar climate ambitions are translated into different contractual arrangements.

The cases are all Swedish and are situated in municipal or national public procurement contexts. This creates contextual coherence, since the procurements operate under similar legal and institutional conditions. At the same time, the variation between municipal projects and a national bonus model allows comparison across different forms of public-sector procurement practice.

*Table 1: Overview of the four cases examined.*

| Case | Procurer | Project type | Instrument design | Interviewees | Period |
|---|---|---|---|---|---|
| **Case 1: Slakthusområdet E02** | Stockholm Stad - Exploateringskontoret | Earthworks for urban-development sub-project E02 | Requirements-based with competitive dialogue (50% electrified machinery and transports) | P1 (procurer); C1 (contractor) | 2023-2024 |
| **Case 2: Trafikverket national bonus system** | Trafikverket (national infrastructure agency) | National policy instrument applicable across infrastructure contracts | Bonus-based per-kWh incentive; mandatory inclusion of bonus clause across new contracts, contractor uptake voluntary | P2 (senior expert, Trafikverket) | Rolling from 2024 |
| **Case 3: Mäster Samuelsgatan / Jakobsbergsgatan** | Stockholm Stad - Trafikkontoret | Inner-city street reconstruction | Penalty-centred with competitive dialogue (electrification thresholds) | P3 (procurer); C3 (contractor) | 2024-2025 |
| **Case 4: Östra Fyrislund Etapp 5-6** | Uppsala Kommun | Urban development and infrastructure project | Requirement-based with fixed high penalty; mandatory electric excavator and minimum energy-share requirement | P4 (procurer); C4 (contractor) | 2024-2025 |

## 4.3 Empirical material

Case study research refers to a research strategy rather than a specific data collection method and may include several techniques such as interviews, document studies, surveys and observations (Priya, 2021). In this thesis, the empirical material consists of three main sources: semi-structured interviews, procurement documents and workshop observations from the SEEK project.

The interviews constitute the primary empirical material of the thesis. They are used to understand how procurers and contractors interpret procurement instruments, risk allocation, participation conditions and implementation feasibility. The procurement documents are used as a complementary source to identify the formal design of the procurement instruments. This includes how electrification requirements, bonus structures, penalties, qualification criteria, verification routines and responsibilities are expressed in written procurement material. The workshop observations are used as contextual material to improve the broader understanding of procurement-model development for electrified construction.

The use of several empirical sources strengthens the study by reducing reliance on one type of material. Interviews provide insight into actor interpretations and practical experiences. Documents provide insight into the formal procurement structure. Workshop observations

provide additional contextual understanding of recurring challenges discussed by actors involved in the broader transition towards electrified construction. Together, these sources make it possible to examine both formal procurement design and actor interpretation.

### 4.3.1 Semi-structured interviews

Semi-structured interviews were used as the primary method of data collection. This method was chosen because the study aims to understand how actors interpret procurement instruments and their implications. Semi-structured interviews are suitable for this purpose because they allow the researcher to ask comparable questions across interviews while still giving respondents room to elaborate on case-specific experiences and introduce themes that may not have been anticipated in advance.

Seven interviews were conducted with actors connected to the four cases. The interviewees consisted of public procurers, a public-sector expert and contractors involved in the selected procurement cases. Procurers were interviewed to understand the intentions behind the procurement instruments, how they perceived the role of the public actor, and how risk and feasibility were considered in the procurement design. Contractors were interviewed to understand how the procurement instruments were interpreted from the supplier side, including how requirements, bonuses, penalties, charging conditions and market maturity affected perceived feasibility and participation conditions.

Two versions of the interview guide were used: one for procurers and one for contractors. The guides covered similar overarching themes, including procurement design, electrification requirements, risk allocation, feasibility, market conditions, incentives, penalties and follow-up. However, the guides were adapted to each case and interviewee role. For example, interviews connected to bonus-based instruments included more detailed questions on bonus levels, investment incentives and voluntary uptake, while interviews connected to requirement-based cases included more detailed questions on compliance, feasibility and risk distribution.

All interviews were conducted in Swedish, recorded and transcribed. The interviews conducted with P1, P3 and P4 were held in person, while the remaining interviews were conducted digitally. Table 2 presents an overview of the interviews included in the study.

*Table 2: Overview of interviews.*

| Code | Actor type | Case | Format | Length | Date |
|---|---|---|---|---|---|
| P1 | Procurer | Slakthusområdet E02 | In person | 55 min | 18 March |
| C1 | Contractor | Slakthusområdet E02 | Digital | 53 min | 30 March |
| P2 | Senior expert | Trafikverket bonus model | Digital | 49 min | 27 March |
| P3 | Procurer | Mäster Samuelsgatan/Jakobsbergsgatan | In person | 50 min | 30 March |
| C3 | Contractor | Mäster Samuelsgatan/Jakobsbergsgatan | Digital | 63 min | 22 April |
| P4 | Procurer | Östra Fyrislund Etapp 5-6 | In person | 53 min | 8 April |
| C4 | Contractor | Östra Fyrislund Etapp 5-6 | Digital | 38 min | 23 April |

The interviews were first transcribed using automated transcription software. Saunders *et al.* (2016) note that transcription can be time-consuming, and automated transcription was therefore used to create an initial transcript. Since automated transcription can contain errors, particularly when technical terminology, Swedish dialects or industry-specific expressions are used, the transcripts were subsequently reviewed and corrected. AI-supported tools were used to assist with cleaning transcript text and organising preliminary thematic material. However, the final interpretation, quote selection and empirical use of the material were manually checked by the author against the transcripts and, where needed, the original audio recordings. This was done to ensure that the empirical material used in the thesis accurately reflected what was said in the interviews.

Since all interviews were conducted in Swedish while the thesis is written in English, interview quotations used in the final thesis were translated. The aim of the translation was to preserve the meaning of the original statements rather than to produce literal word-for-word translations. The original Swedish transcripts were retained during the analysis in order to make it possible to check translated quotations against the original wording.

### 4.3.2 Document analysis

Procurement documents were analysed as a complementary empirical source. The purpose of the document analysis was not to replace the interview material, but to provide a formal baseline for understanding the procurement instruments in each case. While interviews show

how actors interpreted the procurement designs, the documents show how the instruments were formally expressed in procurement and contract material.

The document analysis focused on identifying how electrification-related requirements, incentives, penalties, qualification criteria and follow-up mechanisms were formulated. The documents were also used to clarify how responsibilities were positioned between procurer and contractor, particularly in relation to charging infrastructure, reporting obligations and compliance. This was important because several interview themes concerned not only whether electrification was required or incentivised, but how responsibility for making electrification feasible was distributed.

The document material differed between cases depending on availability. For Slakthusområdet, available material included the administrative requirements covering procurement procedure, contract conditions and penalty structure, the project-specific requirements document detailing machine and fuel specifications and transport reporting, the signed contract including the open-books clause and the sustainability-bonus clause, and the contract appendix specifying the fixed fee structure. For Östra Fyrislund, available material included procurement and contract material describing the procurement procedure, electrification requirements, environmental requirements, penalty structure, qualification requirements and follow-up routines. For Mäster Samuelsgatan and Jakobsbergsgatan, available material included documents related to the competitive dialogue process, qualification requirements, evaluation model, final administrative requirements, contract requirements and penalty structure. For Trafikverket, the available material concerned the bonus model for electrification, including the bonus level, eligible use, reporting requirements and payment logic, supplemented by the IVL background report cited in the bonus model's design. Table 3 summarises the procurement and contract documents analysed across the four cases.

The document analysis was structured around a set of categories derived from the conceptual framework and the research questions. These categories included procurement instrument type, electrification target, qualification effects, bonus or penalty structure, verification method, charging and infrastructure responsibility, and implications for risk allocation. This made it possible to compare how each case formally positioned electrification-related obligations and incentives.

The document analysis served three main functions. First, it supported the description of each case by identifying the formal procurement instrument design. Second, it allowed for comparison between formal contract logic and interviewee interpretations. Third, it strengthened the credibility of the analysis by making it possible to triangulate interview claims against written procurement material. For example, where interviewees described a case as requirement-based, bonus-based or penalty-centred, the documents were used to examine how this was reflected in the actual procurement structure.

*Table 3: Procurement and contract documents analysed across the four cases.*

| Case | Document title | Document type | Date |
|---|---|---|---|
| **Slakthusområdet E02** | Administrativa föreskrifter, handling 06.01 | Administrative requirements, document 06.01 | 2023-10-01 |
| | Särskilda krav för entreprenadens genomförande, handling 01.02 | Project-specific requirements, document 01.02 | n.d. |
| | Entreprenadkontrakt | Signed contract | 2024-05-02 |
| | Kontraktsbilaga 2 | Contract appendix | 2024-05-02 |
| **Östra Fyrislund Etapp 5–6** | Tilldelningsbeslut | Award decision | 2025-01-22 |
| | Kontrakt nr 82840 | Signed contract | 2025-02-05 |
| | Administrativa föreskrifter | Administrative requirements | 2024 |
| **Mäster Samuelsgatan / Jakobsbergsgatan** | Ansökningsinbjudan, handling 1 | Invitation to apply for competitive dialogue, document 1 | 2024-06-28 |
| | Ansökningsdokument (Läskopia), handling 2 | Application document, docuement 2 | 2024-06-28 |
| | Minnesanteckningar, Dialogmöte 1 – Emissionsfrihet | Dialogue meeting notes | n.d. |
| | Administrativa föreskrifter, handling 09.1 | Final administrative requirements, document 09.1 | 2025-03-28 |
| | Utvärdering, handling 09.2 | Final evaluation model, document 09.2 | 2025-03-28 |
| **Trafikverket bonus model** | Bonusmodell för elektrifiering (email correspondence) | National bonus policy text | 2024 |
| | bransch.trafikverket.se | Public web reference | Accessed 2026 |
| | IVL, *Analys av stödsystem för miljöarbetsmaskiner* | Background report | 2025 |

*Note: Document availability differed by case. The three project-specific cases included tender, contract, and administrative documentation in full. The Trafikverket bonus model was studied via email correspondence quoting the relevant procurement-document section (EK §6.133), the publicly accessible bransch.trafikverket.se page, and the IVL background report referenced in the model's design.*

### 4.3.3 Workshop observations

In addition to interviews and documents, observations from SEEK workshops were used as contextual material. The SEEK project concerns procurement and investment models for electrified construction and includes actors from both public and private sides of construction procurement. Four workshops organised by SEEK were attended during the research process, held between February and April 2026, each lasting approximately 2 hours. The workshops took the form of roundtable discussions between stakeholders from construction companies, subcontractors, public procurement units, and researchers connected to the project. The aim of the workshop series was to formulate a recommended procurement model for electrified construction, with individual sessions addressing topics including risk allocation between procurer and contractor, infrastructure responsibility, market maturity, and the design of procurement models that are both ambitious and feasible. These discussions provided further perspectives from actors not directly interviewed and surfaced recurring challenges that the thesis subsequently investigated.

The workshop observations were not treated as the core empirical material of the thesis. Instead, they were used to improve the broader understanding of how electrification procurement is discussed among relevant actors, including procurers and contractors. Notes from the workshops helped contextualise the interview material and supported the understanding of recurring issues such as risk allocation, infrastructure responsibility, market maturity and the difficulty of designing procurement models that are both ambitious and feasible.

Because the workshop observations were not analysed systematically in the same way as the interviews, they are not used as the main basis for empirical claims in the findings chapter. Their role is instead to support contextual triangulation and to inform the interpretation of the interview and document material. This distinction is important because the thesis gives most analytical weight to sources directly connected to the selected cases.

## 4.4 Data analysis

The analysis followed a thematic and comparative logic. The interview transcripts were first read to identify passages relevant to the research purpose and research questions. Relevant passages were then organised according to recurring themes, such as procurement instrument design, bonus logic, requirement logic, penalties, competitive dialogue, risk allocation, investment feasibility, charging infrastructure, verification and participation conditions.

The analysis began with a within-case reading. Eisenhardt (1989) emphasises within-case analysis as an important step before cross-case comparison in case study research. Each case was therefore first examined individually to understand its procurement design, case context and actor interpretations. This was important because the four cases differ in their procurement instruments and project conditions. Analysing each case separately helped preserve the specific logic of each procurement before comparing across cases.

After the within-case analysis, the material was compared across cases. Cross-case comparison was used to identify similarities and differences between procurement

instruments. This made it possible to examine how different designs appeared to position risks and responsibilities differently. For example, requirement-based designs were compared with bonus-based designs in terms of how they placed feasibility risk, investment risk and compliance risk. Cases involving competitive dialogue were also compared with cases where such dialogue was absent or less central.

The document analysis was integrated into this process by coding formal procurement clauses and mechanisms into comparable categories. This allowed the study to compare the formal design of each procurement with how interviewees interpreted its practical effects. In this way, the analysis was not based only on interviewees' perceptions, but also on the procurement documents that structured the conditions for those perceptions.

The thematic analysis was also guided by the conceptual framework developed in Chapter 3. The framework provided the main analytical categories: procurement instrument design, risk allocation and responsibility positioning, actor size and supply-chain position, participation conditions and implementation feasibility. However, the analysis was not limited to these categories. Themes that emerged from the empirical material, such as competitive dialogue as a learning mechanism and verification as an unusually visible aspect of electrification requirements, were also incorporated into the findings.

The result of the analysis is presented thematically in the findings chapter. The themes were selected because they recur across several cases or because they reveal important contrasts between cases. The findings chapter therefore does not present each case in isolation, but uses the cases comparatively to answer the research questions.

*Table 4: Analytical categories and example codes used in the thematic analysis.*

| Analytical category | Example codes |
|---|---|
| **Procurement instrument design** | mandatory requirement, bonus, penalty, combinations |
| **Risk allocation** | utilisation risk, financing/CAPEX risk, compliance risk, enabling-condition risk |
| **Actor size and supply-chain position** | small contractor, large contractor, machine owner, subcontractor |
| **Participation conditions** | qualification criteria, tendering uncertainty, access to equipment, lead time |
| **Implementation feasibility** | charging access, infrastructure provision, preparation time, productivity effects |
| **Cross-cutting mechanisms** | competitive dialogue, market dialogue, follow-up routines |
| **Verification** | site inspection, kWh reporting, monthly reporting, digital follow-up |
| **Anchoring** | political mandate, organisational policy, individual initiative |

## 4.5 Use of AI-supported tools

This section discloses how AI-supported tools were used during the thesis work. Two general-purpose AI-supported tools were used during different stages: ChatGPT (OpenAI), primarily in the earlier phase, and Claude (Anthropic), primarily in the later phase. The tools

were used as aids for reflection, structuring, language and document preparation, while research design, empirical material, analysis and conclusions remained the responsibility of the author.

In the early phase, AI tools were used to support the framing of the study. This included discussing and refining the research questions as the empirical scope developed, reasoning about how the four cases could be compared. AI tools were also used as a sounding board when structuring the conceptual framework in Chapter 3, including organisation of risk categories and cross-cutting mechanisms. The underlying arguments, source selection and theoretical positioning were determined by the author.

During data collection and analysis, AI tools were used to assist with practical and organising tasks. They supported the drafting and adaptation of interview guides, while the interviews themselves were planned and conducted by the author. AI tools provided initial cleaning of transcripts and initial pattern recognition for themes, which the author then manually corrected. All coding, theme development, quote selection and interpretation were carried out and verified by the author against the transcripts and, where needed, the original recordings. The pseudonymisation of interviewees into role-based codes was also supported by the AI tools and checked by the author to confirm that the pseudonym was correctly attributed to each finding and quote.

In the writing phase, AI tools were used for language editing, translation and document preparation. This included support with initial translation of interview quotations from Swedish to English with the aim of preserving the meaning, producing the Swedish summary, improving clarity and flow, condensing text, and structuring chapters such as the findings and discussion. AI tools were also used to generate the formatted tables and the conceptual-framework figure. Reference management was carried out by the author using Zotero, and all cited sources were verified by the author.

Throughout the project, AI tools were also used to support the planning and management of the work itself. This included breaking the work into prioritised tasks and checklists in order to meet deadlines, using AI as a sounding board to check whether revisions addressed feedback from the supervisor and seminars, additionally it was used to review if changes preserved the overall coherence and argument of the thesis. This use supported the organisation and progress of the work rather than its analytical substance.

## 4.6 Research quality and triangulation

Because this is a qualitative case study, the quality of the research is not primarily assessed through statistical validity or generalisability. Instead, the study aims for analytical transparency, credible interpretation and clear connection between empirical material and theoretical argument.

Baxter and Jack (2008) emphasise the relevance of using several data sources in qualitative case study research. In this thesis, triangulation was used to strengthen the credibility of the study. Interviews provided insight into actor interpretations, while procurement documents provided insight into formal procurement design. Workshop observations provided contextual

understanding of the wider procurement-development environment. These sources were not used in the same way, but they helped support each other. Where possible, claims made in interviews were compared with procurement documents to determine whether the formal structure of the instrument supported or complicated the interviewee's interpretation.

The use of both procurer and contractor interviews also contributed to the quality of the analysis. Procurers and contractors occupy different positions in the procurement process and therefore interpret risks and feasibility from different perspectives. Including both sides made it possible to compare not only different instrument designs, but also different actor interpretations of the same or similar procurement mechanisms.

The study does not aim to generalise statistically to all Swedish public procurement or all construction projects. Instead, it aims to contribute analytical insights into how procurement instrument design can structure risk allocation and feasibility in electrification-oriented construction procurement. The findings may therefore be relevant for similar contexts where public actors use procurement to support emerging climate technologies, but the conclusions should be interpreted in relation to the cases and context studied.

## 4.7 Ethical considerations

Ethical considerations were important because the study relies on interviews with actors discussing procurement strategies, project experiences and potentially commercially sensitive issues. Before each interview, the participants were informed about the purpose of the study, how the material would be used, and that their participation was voluntary. Consent was obtained for recording the interviews.

The interviewees were anonymised through role-based codes such as P1 and C1. This was done to protect respondents while still making it possible for the reader to understand which type of actor and case each statement relates to. Since the cases themselves are identifiable, full anonymity cannot be guaranteed in an absolute sense. However, the thesis avoids unnecessary personal details and does not attribute statements to named individuals.

The handling of documents was also considered from an ethical perspective. Procurement documents that are public or provided for research purposes were used to support the analysis, but sensitive or confidential information was not included where it was not necessary for answering the research questions. The study follows the general principles of informed consent, confidentiality and responsible handling of research material as described by Vetenskapsrådet (*Good Research Practice 2024*, 2024).

## 4.8 Methodological limitations

There are several limitations to the methodology. The first limitation concerns the number of interviews. Seven interviews provide valuable insight into the selected cases, but a larger number of interviews could have provided a broader understanding of each stakeholder group. This is especially relevant because procurement for electrified construction involves several actors beyond the procurer and main contractor, including subcontractors, machine owners, operators, OEMs, charging-infrastructure actors and grid companies.

A second limitation is that the study does not include direct interviews with smaller subcontractors or machine owners. This is important because the conceptual framework highlights that actor size and supply-chain position may influence the ability to absorb risk and participate in electrification-oriented procurement. Since these actors are not directly represented in the interview material, the thesis cannot make strong empirical claims about their own experiences. Instead, the analysis remains attentive to how risks may cascade through the supply chain based on the perspectives of procurers and main contractors, supported by procurement documents and workshop observations.

A third limitation concerns the comparability between cases. The cases were selected because they represent different procurement instrument designs, but they also differ in project type, procuring organisation, timing, market conditions and contractual details. This means that differences between cases cannot be attributed solely to the procurement instrument used. The analysis therefore treats instrument design as a central analytical dimension, but not as the only factor shaping participation and feasibility.

A fourth limitation relates to the novelty and rapid development of electrified construction. The availability of electric machines, charging solutions and contractor experience is changing quickly. Findings about feasibility and market maturity should therefore be understood in relation to the time period in which the cases were conducted. A procurement design that was perceived as ambitious or risky during one case may be perceived differently as the market develops.

A fifth limitation concerns the use of interviews as the primary empirical material. Interviewees may describe events from the perspective of their organisation, may emphasise successful aspects of a project, or may be cautious when discussing problems in procurement processes. This limitation was addressed by comparing interviews with procurement documents where possible and by including both procurer and contractor perspectives. However, the study remains interpretive and focuses on perceived implications rather than measured project outcomes.

Despite these limitations, the methodology is appropriate for the purpose of the thesis. The combination of interviews, document analysis and contextual workshop observations provides a basis for analysing how procurement instruments are designed, interpreted and connected to risk allocation, actor participation and implementation feasibility in electrification-oriented construction procurement.

# 5. Findings

This chapter presents the empirical findings from the four cases examined in the thesis: Slakthusområdet E02, Trafikverket's bonus model, Mäster Samuelsgatan and Jakobsbergsgatan, and Östra Fyrislund Etapp 5-6. The chapter is organised thematically. Section 5.1 examines how electrification is anchored through mandates, individual initiative and institutional learning. Section 5.2 analyses how procurement instruments distribute risk between procurers and contractors. Section 5.3 examines conditions for participation across

the value chain, including demand signals, market maturity, charging infrastructure and subcontractor coordination. Section 5.4 focuses on verification and follow-up. Together, the sections show how procurement design affects the balance between climate ambition, actor participation and implementation feasibility.

Consistent with the exploratory design of the study, the findings are organised around the themes that emerged across the cases rather than around the research questions. Sections 5.2 and 5.3 correspond most directly to the first and second sub-questions, on risk positioning and on participation conditions. Section 5.4, on verification and follow-up, informs both the design of the instruments (Sub-Q1) and the tension between ambition and feasibility (Sub-Q3). Section 5.1, on the anchoring of electrification ambition, emerged as a cross-cutting theme that is not tied to a single sub-question and is developed further in the discussion and conclusion. The discussion in Chapter 6 then revisits these findings through the structure of the research questions, in order to answer them directly and build towards the main research question.

To support readability, interviewees are referred to using role-based codes throughout the findings chapter. “P” refers to procurer-side or public sector respondents, while “C” refers to contractor-side respondents. The number corresponds to each case in the order they appear above. These codes refer to individual interviewees rather than to the full organisation they represent.

*Table 5: Interview codes used in the thesis.*

| Code | Actor perspective | Case |
|---|---|---|
| P1 | Procurer-side respondent | Slakthusområdet E02 |
| C1 | Contractor-side respondent | Slakthusområdet E02 |
| P2 | Public-sector expert | Trafikverket bonus model |
| P3 | Procurer-side respondent | Mäster Samuelsgatan/Jakobsbergsgatan |
| C3 | Contractor-side respondent | Mäster Samuelsgatan/Jakobsbergsgatan |
| P4 | Procurer-side respondent | Östra Fyrislund Etapp 5-6 |
| C4 | Contractor-side respondent | Östra Fyrislund Etapp 5-6 |

## 5.1 Anchoring electrification ambition

The concept of electrified construction projects is not a completely natural evolution. The empirical material shows that in each case there has been a commitment from the procuring organisation that is the basis for implementing the different procurement tools with the

ambition of getting a partial or fully electrified project. This internal commitment is needed in order to facilitate the electrification of construction projects since it still entails overall higher costs for the procuring party. The motives that lay the foundation for this investment differ between the investigated cases. Two of the cases, Slakthusområdet E02 and Östra Fyrislund etapp 5-6, base their motives for including electrification requirements on stated political mandates. The Trafikverket case is grounded in organisational policy, with an electrification bonus widened to include all future projects. The case of Mäster Samuelsgatan introduces a different variant: the individual interest and ambition of the procurer responsible for the project.

> *"It seemed very natural that we continue with that concept (electrification) [...] it is stated in our budget directives, we are to work with such things more generally." – P1*

> *"From now on, the bonus is mandatory in all contracts. It is no longer up to our project managers to choose." – P2*

This indicates that political mandates and set policy objectives are a common rationale behind electrified construction projects. Although the example of individual patronage in the Mäster Samuelsgatan procurement shows that there can be different paths to initiating a transition in which electrified aspects are included in construction projects. Overall, there was consensus across the interviews in the need for some type of motive to make this transition towards electrified construction projects a reality, in whichever form it takes.

Looking at the institutional push for electrification there can be a general observation of a trend with politicians coming with lofty goals about climate action which then needs to be adapted by employees in the public sector. Sometimes it is pre-determined and specified with set goals as can be seen in the Östra Fyrislund case, here the procurer describes how there is a political decision that 50 % of the procured vehicle fleet should be electric by 2030. Logically this would entail a gradual increase in their operations until then as to properly be able to access the vehicles and create the market for companies to fill. P4 describes how, due to differences in types of contracts the municipality procures, different types of procured projects might need to go beyond 50 % electrified vehicles due to other project having worse conditions to meet their quota, since the 50 % is related to the overall procured fleet (not only in construction). There is also the general interpretation by procurers that increased electrification is what is desired by the local politicians, even when not directly stated, this lends space for the individual procurer to interpret general guidelines rather than following strict plans.

Individual championship is an alternative driving factor behind increased electrification commitment in publicly procured construction. In the Mäster Samuelsgatan case, the procurer took a leading role in the initial stages, aiming to make the project fully electrified out of her own interest in working with electrified construction. Additionally, the project revolved around an area designated to be an environmental zone class 3, where only electric vehicles are to be permitted, and as such P3 deemed it fitting to put the same requirements on the construction project. P3 further explains how she had to recruit colleagues and convince

decisionmakers in the office as to realise this vision. Ultimately, P3 got the green light with the prerequisite of keeping to the pre-determined budget. Furthermore, P3 specifically highlights the lack of clear directives from politics in this case.

> *"We have not had any implementation decision from the political level saying that we should carry it out electrified, rather it is something we ourselves have pushed for." – P3*

The motives revolved around learning more about the intricacies of electrified construction projects and this personal motivation led to the establishment electrification criterion in this case. This scenario highlights the potential the individual procurer has regarding initiating initiatives and creating the conditions for an electrified construction project where before there was no plan for it.

Individual championship and motivated personnel can be key drivers in finding potential for electrification outside clear directives. The procurer for Slakthusområdet reflects similarly, noting that certain people in the office are actively engaging in the question of electrified construction projects due to personal interest and that they feel that it is important. Additionally, P1 describes how the project at Slakthusområdet was politically mandated to be emissions-free, but he highlights the distinction from electrified, with the electrification stemming from the procurers interpretation of a vague political goal.

> *"There are a number of people who want to drive these issues, who think it is important and who find it engaging." – P1*

This illustrates how personal engagement among individual officers, rather than formal direction alone, can sustain electrification ambition within an organisation.

The size of the procuring organisation is an additional factor to consider on the path towards more electrified construction since larger municipalities can be better situated to lead the way. The procurer for Östra Fyrislund supports this view. He reasons that larger actors such as larger municipalities and government agencies need to commence the transition and explore which pathways to take in electrified construction projects; even that it is their responsibility. This stems from the large costs and the time investment needed to make the transition, something smaller procuring parties cannot afford. Simultaneously he reasons that larger actors hold more sway with the entrepreneurs as their requests carry more weight.

This difference in internal knowledge and organisational capacity is also reflected in the formal choice of procurement procedure. The choice of procurement procedure differs across the cases and is reflected in the formal procurement documents. Östra Fyrislund Etapp 5-6 was conducted as an open procedure under chapter 19 Swedish Public Procurement Act, with award based solely on lowest price and electrification as a mandatory contract condition (tilldelningsbeslut 2025-01-22, kontrakt nr 82840). The documents show that the lowest-priced bidder, was disqualified for failing to meet the obligatory requirement for an electric excavator, illustrating how contract-stage electrification requirements can determine market outcomes even in a price-only procurement. Slakthusområdet E02 and Mäster Samuelsgatan

and Jakobsbergsgatan, by contrast, used competitive dialogue (KPD). The invitation to apply document for Mäster Samuelsgatan (dnr T2024-01813, dated 2024-06-28) explicitly motivates this choice on the grounds that a successful project requires "innovative handling of both contract and technology" and that the city wanted to "capture market knowledge." The procedural choice thus reflects a documented institutional stance: where procurer knowledge is sufficient, an open procedure with strict requirements is used; where it is not, competitive dialogue is deployed before the contract terms are fixed.

Competitive dialogue appears in two of the cases, Slakthusområdet and Mäster Samuelsgatan. In both, the procedure is deployed before the contract is finalised, allowing the procuring agency to engage OEMs, contractors, and subcontractors in scoping feasibility before the requirements and contract structure are set. Across the interviews three analytical functions of dialogue emerge: integrating market knowledge into the procurement, bounding risk before contractual commitment, and accelerating institutional learning that can eventually make the procedure redundant.

In the case of Slakthusområdet P1 described competitive dialogue as a means to improve reach within the value chain, with OEMs and subcontractors being able to provide understanding of capabilities before the contract is specified. The rapid pace experienced within electric machine capabilities and supporting technologies was deemed to be a knowledge barrier. Through early involvement of contractors the boundaries of what feasibility could be achieved within the project could more easily be determined, as knowledge from different actors could be incorporated when creating the contract clauses. The procurer for the Mäster Samuelsgatan case shares this perspective, describing how the lack of previous experience on procurement for electrified construction projects made the dialogue essential for understanding what the market could deliver. When commencing the process, P3 describes sending out invitations to relevant actors with the qualifications they are looking for, in this case described as quite basic, namely some experience of electrification in projects. C3 shares this perspective on competitive dialogue and highlight that it enabled them to discuss the feasibility of 100 % electrification in this project while making the procurer aware of the impact it would have on costs.

> *"I think it was absolutely crucial for it to be feasible to carry out, since the client did not know how to phrase the question and they did not know what they could demand. So this dialogue allowed us to tell them how it works, what we can deliver, and how they should phrase the question." – C3*

The contractor at Östra Fyrislund, a case where competitive dialogue was not used, describes an uncertainty relating to the electrification aspects and requirements in the contract. The lack of experience on electrified projects from the contractor side in combination with unclear or difficult to understand contract clauses has the potential to increase the perception of risk, especially since this case had a high penalty clause connected to electrification. C4 argues that this probably made many tenders more expensive since they compensate for the potential fines they risked receiving.

Beyond integrating immediate market knowledge, competitive dialogue also builds longer-term relationships between procurer and contractor. A further reflection brought forward by P3 is the dialogue's ability to improve the connection to the different actors, offering an opportunity for improved relationships between public and private actors. This indicates that there can be a further advantage related to gaining a general understanding of the market, what entrepreneurs are available, how they work, and at what level they can deliver for future projects. The main finding here is that the time it takes to create a good list of requirements in a scenario of changing conditions can be lessened by involving expertise outside of the procuring organisation.

> *"Since we did not have this kind of knowledge about electrification from before [...] we thought it was a good idea to try a procedure called competitive dialogue." – P3*

Regarding the selection of applicants, the dialogue is perceived as inconsequential since there are initial pre-requisites for entering the dialogue process, already eliminating actors at this point according to P1. He adds that the normal criteria set in a procurement contract are effective enough to eliminate insufficient tenders. Knowledge integration, rather than applicant selection, is therefore the mechanism through which dialogue adds value in these cases.

The formal qualification documents for Mäster Samuelsgatan supports this picture by showing that in that case, electrification experience was explicitly used as a selection criterion rather than being left to the dialogue alone. The invitation to apply document (document 1, 2024-06-28) and the qualification requirements (document 2, section 4.5) specify demonstrated experience of emission-free machinery both at company level and at the level of the named site manager, with the latter required to evidence this through up to three reference projects. In the merit evaluation (document 1, section 5), emission and fossil-free competence within construction projects was weighted at up to 6 of 12 possible points, the single highest-weighted selection criterion. Electrification experience therefore functioned both as a market signal to contractors and as a gate that shaped which actors were invited into the dialogue at all. The tension P1 observes, that pre-qualification criteria can eliminate actors before the dialogue even begins, is thus visible in the documents: the Mäster Samuelsgatan KPD process selected for actors with emission-free experience precisely because the procurer judged that actors without it could not contribute meaningfully to a dialogue about full electrification.

An alternative perspective on competitive dialogue was brought forward in the interview relating to the case Östra Fyrislund. Here P4 is aware of the process and that it has been used in Stockholm, although decides not to use it, preferring instead to set strict requirements on electrification. Although the contractors were initially worried about feasibility of the task, the procurer argued that by giving adequate preparation time, in this case about 9 months between the signing of the contract and start of the project, the contractors could manage to access the proper equipment and manage electrification without the need for competitive dialogue. C4 agrees that the time is crucial when preparing for an electrified project due to the increased difficulty of accessing electric machines, stating that in this project they had

adequate time. P4 later talks about the importance of being educated on the subject when dealing with contractors as they might try to exaggerate difficulties and lower expectations on electrification levels in projects. This should be considered so as to not relinquish too much power to the contractor in a competitive dialogue.

The use of competitive dialogue is not purely advantageous as it can force a lock-in towards specific contractors and damage competition. In the specific case of Slakthusområdet the procurer describes a situation where the dialogue with the contractor changed the nature of procuring. The budget was presented from the procuring party and the discussion then revolved around how low the CO2 emissions the contractor could achieve based on that budget. A further observation brought up in this interview was the ability to better prepare the contractor before the contract had been signed, allowing the contractor to reach their subcontractors and involve them in the discussion of what could be achieved. The consequence of this is described as becoming more locked in towards these contractors before a price is agreed upon. This could indicate that the process hampers competition between actors and in turn undermines the value-for-money aspect of the procurement although the improved ability stemming from an earlier involvement of sub-contractors offer its own advantages.

One of the main strengths of competitive dialogue is its ability to shed light on what is possible under the conditions of the project before the contract is set, and what impact certain decisions might have on the length and cost of the project. Examples from the Slakthusområdet interview include the ability to get a price for a machine and identify what conditions apply for that price to be sustainable for the contractor. The potential for optimisation and for reducing the risks relating to electrification can thus be discussed before either party needs to commit to a plan of action.

> *"So it is largely about finding the optimisation in order to eliminate the uncertainties that may be cost-driving; that is largely the benefit of the dialogue." – P1*

Competitive dialogue gives the contractor an opportunity to calibrate procurer expectations and surface feasibility concerns before the contract is set. From the perspective of C1 the competitive dialogue was seen as a valuable process as it offered an ability to temper expectations from the procuring party and make the contract feasible, especially important when working in more complex projects such as electrification transitions. C1 further describes that in other scenarios there had been situations where requirements were out of step with reality and the procurer did not know what certain aspects entailed. This indicates that requirements may be poorly calibrated in scenarios without a competitive dialogue, although this could potentially be managed through other means, such as more knowledgeable procurers or a strong existing relationship between contractor and client. Exemplifying this, the contractor in the Mäster Samuelsgatan project reflects that through the collaboration between contractor and procurer the goal of 100 % electrification could be verified. An important aspect in this case, where the involved parties lacked experience in electrified construction, as the goal can be set but verification of the feasibility and consequent costs need to be understood before commencing the project.

Another aspect brought forward by the contractor at Mäster Samuelsgatan is that the competitive dialogue offers the possibility for the contractor to impact the final formulation of the contract. C3 describes how he championed for the strong fines, connected to the electrification level, included in the contract as to discourage planned contract violation from tendering entrepreneurs. Additionally, C3 managed to add the qualitative assessment, with price discounts, as a part of the tender evaluation to benefit tenders with true ambitions on electrification, in contrast to a pure lowest-price evaluation. This shows how the competitive dialogue can be a resource both for the procuring party but also for the contractor as they get the opportunity to steer the contract. In situations such as this project it meant that important measures protecting the electrification ambition of the project were included in the contract although procurers should be vigilant as to not completely relinquish control of the contract to the opposite party as this could entail worsened conditions for the intended electrification or increased price if the contractor is more malevolent. C4 offers a different perspective on quality-centred procurement, reflecting on that it might be a good path although she argues that lowest price still should win. She is especially critical of reference projects and experience of electric projects since that would negatively impact competition, excluding most other actors when so few project have yet been conducted electrified.

At the same time, there are situations where competitive dialogue is redundant, specifically when the perceived knowledge of the subject is already entrenched in the procuring organisation. The procurer for Slakthusområdet elaborates by describing how competitive dialogue has been used in further cases where it has allowed for early recognition of the feasibility of implementing electrification aspects. This indicates that the use of the tool helps procurers identify what expectations could be held when evaluating criteria for procurement contracts in an evolving or novel market. This situation was described by P1 when recounting a following case where they had gained enough experience to revert to a traditional process for determining contract details. This perspective was also reflected in the Mäster Samuelsgatan case, where after using competitive dialogue to create a proper understanding of electrified projects in several cases, the procurer decided to revert to a more traditional methodology for following electrified projects. She reasons that if the Mäster Samuelsgatan project were to be commenced today she would not use competitive dialogue.

> *“Now we have so much knowledge that, had I been in the same situation again, I would not have needed to carry out a competitive dialogue [...]” – P3*

When the procurer deems they have an adequate understanding of the intricacies of the project, electrification in this context, they might opt not to go through with competitive dialogue since it does not in itself lead to the desired outcomes once the project is underway.

> *“We have now learned so much about electrification that we do not need to have a competitive dialogue [...]” – P1*

The Östra Fyrislund interview also indicated that the main advantage in formulating these procurement contracts is related to knowledge on the subject of electrification, something P4 felt he had managed to gather from activities such as conferences, fairs and workshops with

industry actors. This supports the idea that once the knowledge is acquired, be it conducting a competitive dialogue or other pathways, this process loses one key advantage for future projects. This further indicates that competitive dialogue mainly is seen as a tool for bolstering competence on a relevant subject area. It was not used as a default option but rather as a compromise when dealing with complex or novel circumstances. The lack of previous market dialogue was mentioned by the procurer for Mäster Samuelsgatan as a reason for taking this path, mainly to evaluate the maturity of the market. This perspective is shared by the contractor counterpart in this project who reflects on the fact that the client side needed support, facilitated by the competitive dialogue, in order to formulate the request in the correct way. The Slakthusområdet procurer reasons similarly noting that they are at a disadvantage when trying to fabricate the specifications in the contract, especially due to the rapid pace of development in regard to electrification.

> *"We do not have time to do it thoroughly, and even if we take the time, it is questionable whether we will be that good at it in relation to how rapid the market develops." – P1*

The procurer at Slakthusområdet reasons that although they used competitive dialogue in that case, subsequently they have moved away from using it and focused on qualitative aspects in the procurement, such as selection on innovation and competence. Additionally, he states that procuring by purely lowest-cost basis rarely is the correct move. This is something the contractor at the Mäster Samuelsgatan case also was a big proponent of, mentioning that qualification on additional factors such as competence and the quality of the tender should be more central if the procurement aims to achieve their objectives, especially in complex projects such as electrification. Otherwise, there is the risk that a lowest cost tender disregards or underestimates the challenges that might come up and as such offers a price that is not sustainable.

## 5.2 Instrument architecture and the distribution of risk

Building on the conceptual framework's notion of instrument architecture, this section examines how the configuration in each case positions risk across the actors involved. Risk is intrinsic to any construction project, and especially so when projects involve a transition to immature technology. When several actors are involved, the question becomes who carries which risks and to what degree, recognising that different actors have varying ability to absorb different forms of risk. No single actor can carry the entirety of the risk in electrification procurement, since the technology is immature and its cost, performance, and operational implications are not yet fully understood by any party. The procurer in the Slakthusområdet case highlights the perceived risk related to cost increases when moving towards more electrification in construction projects. There is an uncertainty on how expensive it is to achieve sustainability results of different degrees in these projects and as such there is an advantage in comparing ideas and offers from different actors as to see what benefit can be achieved at what price. Further reflections relate to the level of risk put on the contractor, where P1 reasons that no matter the size of the actor they cannot afford to electrify if they carry all the risk, something directly corroborated by C4. Therefore, the contract needs

to be dynamic and allow for the client to carry some degree of risk as to improve feasibility for all actors involved. P1 extends this logic to include the OEMs who already have invested heavily into producing this new electric construction equipment and that they inherently have an interest in making the transition feasible to recoup that investment. This could be interpreted as a need for more competitive pricing of these machines from the OEMs as a possible path for spreading the risk along the value chain. P1 acknowledges that it sometimes is possible to transfer additional risk to the contractor although that this would logically come with an increase in their price. In this scenario it is important to reflect on which option would be cheaper, with most scenarios being more cost effective when the municipality carries some risk according to P1. This perspective is corroborated by P2 who also sees issues with putting too much risk on the contractor, stating that it will lead to accumulated costs, hindering rather than helping the transition.

> *"Development Administration, here you go, here are 60 million. Please use them to cover the additional cost [...] Do not return the money at the end of the year." – P1*

Spreading risk across an organisation is also an important aspect of risk management. The Slakthusområdet procurer reflects that the municipality can absorb risk by spreading it across many contracts, since their largest contractor, Skanska, represents only around 20 percent of their total spend. The remaining 80 percent of their procurement portfolio is distributed across other contractors, giving the municipality room to carry risk that Skanska, with the same project occupying a much larger share of its active workload, cannot. He also describes the importance of reflecting on which party has the better economic position and what interest rates apply. In Stockholm's case, the municipality's strong credit rating means it can bear more financial risk than smaller actors operating with thinner margins and higher capital costs.

Risk in immature-technology procurement does not behave like traditional construction risk. The Slakthusområdet procurer argues that over-burdening the contractor with this risk produces what he describes as a "huge sub-optimisation," with any form of electrification becoming too expensive when risks are poorly understood and the contractor carries too many of them. The issue is therefore one of procurement structure rather than the cost of electric construction equipment itself.

Looking at bonuses, they have the potential to function as a risk subsidy easing the risk burden on contractors without forcing the transition. This is described in the Trafikverket case where the interviewee argues that the bonus model with incitements for electrification rather than requirements is a more risk averse way of working, stating that if you put strict requirements in emerging markets where delivery might be uncertain you risk too high costs and limited contractors tendering for the project. He continues by admitting that this is relevant in the short term although once the market matures bonuses might have overplayed its role. Trafikverket as a public authority has offered bonuses for electrified machines for certain projects previously, and less polluting combustion engine models before that, the difference now is that the bonus now is an overarching rule for all Trafikverket projects. P2

describes the bonus as a risk subsidy, with Trafikverket stepping in and shouldering some of the risk through the payment of this bonus for those contractors who choose to transition.

> *"You could call it a risk contribution […] If they make the decision to acquire an electrically powered machine and take on that risk, then the Swedish Transport Administration steps in with this financial compensation, we shoulder part of the risk." – P2*

The optional nature of the bonus pathway also allows for contractors to assess the feasibility of electrified vehicles for each project, eliminating potential logistical and charging associated risk by not forcing electrification in projects where it is ill-suited according to P2. This directly relates to a previous finding, the risk of incurring drastically increased cost by forcing strict electrification requirements in projects where it is not feasible and directly circumvents it through the optionality of the bonus system.

Procurers face a choice in deciding between setting requirements or incentives when trying to affect change, each has its own advantages and trade-offs. The Trafikverket representative articulates this trade-off in similar terms. He argues that the drawback with using requirements, especially in an emerging market where capacity for delivery is uncertain, since it risks leaving you without any contractors being able to tender and might escalate costs if forced through. With an incentives model like the national bonus, it does not limit participation as everyone can tender as usual although actors interested in making the change get the bonus if they choose to transition to electric equipment he argues. This leaves the decision to the contractor where they can choose whether to invest or not. Although this is an amicable route to take it is a soft approach which might not deliver the intended results due to its voluntary nature, especially since resistance to change is a natural occurrence in most organisations. The contractor at Slakthusområdet believes that a potential bonus needs to be substantial for contractors to commit to the change. The procurer for Östra Fyrislund describes another procurement where they used a bonus, but few parties were interested and they moved towards a requirement focused procurement process to get the results they wanted. He reflects on that requirement, with high fines for mishandling the electrification in this case, is more of a whip rather than a carrot which resulted in some negative responses in the procurement process, but it ultimately got them the results. C4 adds a contractor-side perspective in this tension, being critical towards the use of fines, instead arguing for the carrot as a means to reach the goals. Additionally, C4 suggests that the penalty did not only function as an enforcement mechanism but also became a pricing uncertainty during tendering.

> *"I think that most of those who submitted bids in that process built in some margin for this penalty, because it was difficult to assess at such an early stage what it would entail." – C4*

The contractor perspective on penalties was not uniformly negative. In Mäster Samuelsgatan, C3 instead argued that strong penalties could protect serious bidders by preventing

competitors from promising electrification during tendering and then avoiding it during implementation.

> *"I pushed quite hard for them to have penalties […] So that a contractor cannot promise things and then disregard them once they have the project." – C3*

Requirements and bonuses are not necessarily mutually exclusive; there are opportunities in combining these instruments. Regarding a bonus P4 sees particular potential for bonuses layered on top of requirements when contract change is involved (additional works, ÄTA) or where the contractor can offer electrification beyond the initial requirements of the contract. A combination of strict requirements for the base level with bonuses being paid for delivering further electrified machine hours. Here P4 states that inspiration stems from Trafikverket's bonus although with greater bonus amounts as he sees increased reward to better cover the added costs of electric machines as critical. He sees bonuses as a good method although he is critical of its effectiveness.

> *"I would put it this way: I advocate the bonus arrangement, but with a bonus arrangement we will never reach the targets." – P4*

Across the interviews, bonuses were not rejected as instruments, but their effectiveness was consistently linked to calibration. P2 described the bonus as a way for the procurer to carry part of the investment risk. P4 supported the principle of bonuses but argued that they would not reach climate targets unless the bonus was large enough to change behaviour. C3 similarly viewed bonuses positively when tied to quality or environmental performance, while C1 emphasised that the amount must be substantial enough to offset the additional cost of electric equipment.

Ultimately the bonus might have a role in projects where strict requirements are not feasible due to project-specific conditions, offering an incentive where degrees of requirement are difficult to predict.

Procurers and contractors perceive the risks of electrification differently. Where procurers tend to frame the risk in terms of technological uncertainty and cost premiums, contractors locate the risk more in the mechanics of pricing and project planning under altered conditions. The contractor for Mäster Samuelsgatan acknowledges the additional cost of electrified equipment but states that it inherently does not constitute a risk since they know the prices and adjust their pricing towards the client accordingly. The main issue is the change in project planning, where the logic is reversed compared to a traditional project, it is important to identify what resources are available and then work backwards from there according to C3. He also reasons that if they were to commit to electric machinery in a project where it was not strictly demanded they would be paying double for a machine and not getting paid for it. As such there is a risk connected with substitution to electric machinery when it is outside the scope of the contract, resulting in an increased importance of electrification requirements in contracts as to support this transition, admitting that it will cost more but assist with growing the market. C4 also emphasises the importance of fair cost

compensation for electric machinery, especially in cases where they are less productive and the contractor might need to spend more time using them. Since it is common to rent equipment by the hour and if the contractor needs to spend more time completing a task with electric equipment prices risk escalating quickly if this is not properly understood and accounted for. C4 sees this as a possible knowledge gap from the procurers who need to understand these aspects for them to properly reimburse the contractor or for when designing bonuses.

> *"It costs four times as much to have such a (machine) in the project as a (conventional one). And here I do not really know how the municipality obtains that kind of information." – C4*

In the Slakthusområdet project they had a target price compensation model which made the economic risk minimal as that means they get paid for everything they do, C1 describes it as the client taking on 99 % of the risk. This stems from the fact that it was a pilot project, altering how the contract situation was formulated with focus on technical performance and climate goals rather than price. Going forward C1 argues that risks need to be more evenly distributed otherwise it would not be sustainable.

The contract documents for Slakthusområdet E02 give this account precise institutional grounding. The administrative requirements (AFB.14) specify running account against a target cost as the compensation model, with a fixed fee of 10 percent of the target cost covering Skanska's profit, interest and central overhead (Entreprenadkontrakt, kontraktsbilaga 2). Under §6.2 of the signed contract, all verified self-costs are reimbursed to the contractor with open-book access for the procurer, meaning the economic uncertainty of electrification is absorbed by the municipality rather than priced into the tender. What the contract simultaneously imposes on the contractor is a binding CO2e ceiling: the offered climate calculation (AFB.53) constitutes a cap on total emissions for Phase 2, backed by a stepped penalty of up to 2,000,000 SEK for exceedance (AFD.518). Economic risk and climate risk are thus allocated to different parties, the municipality carries cost uncertainty while Skanska carries performance risk. This is the structural basis for P1's account and makes Slakthusområdet analytically distinct: rather than a single risk allocation decision, the contract design separates risk by type and assigns each to the actor better positioned to manage it.

Overall, risk distribution is strongly shaped by instrument choice, although the cases show that this distribution is modified by compensation models, dialogue, infrastructure responsibility and verification routines. Under bonus-based instruments, contractors decide whether to invest in electrification while procurers carry partial cost risk through the bonus payment, as P2 from Trafikverket emphasised. Under requirements-based instruments the relationship inverts: procurers absorb feasibility risk by mandating outcomes (P4, P1), while contractors bear compliance risk and the operational rework needed to deliver (C3). P1 adds a nuance worth foregrounding, namely that when requirements are paired with dialogue, risk allocation becomes negotiable rather than fixed, suggesting that pure-requirements regimes overstate the rigidity of the procurer to contractor split.

The procurement documents make this risk-distribution logic visible in the formal architecture of each case. Instrument architectures described by interviewees are directly legible in the procurement documents. Östra Fyrislund's administrative requirements (AFC.2231) specify that at least one electric excavator of 12 tonnes or more must be used and that electric-machine operation must account for at least 30 percent of total energy use in the contract works, backed by a fixed penalty of 2,000,000 SEK if the requirement is not met (AFC.512). The contract uses a fixed-price model without index adjustment (AFA.5), meaning the contractor carries all cost risk including any cost premium from electric machinery. Trafikverket's bonus model (EK §6.133) operates on the opposite logic: a voluntary payment of 2.5 SEK per kWh used by qualifying zero-emission machines and vehicles, up to a maximum of 200,000 SEK per contract, with payment conditional on digital verification via BEAst Supply and Peppol. The TCO calculation behind the bonus level, based on IVL analysis showing a monthly cost difference of approximately 8,000 SEK between electric and diesel excavators at 4,000 kWh per month (Bäckström and Grahn, 2025), indicates that the bonus is calibrated to roughly offset the cost premium, consistent with P2's description of the bonus as a risk subsidy. Mäster Samuelsgatan's evaluation model (document 09.2, dated 2025-03-28) monetises electrification ambition directly in the award: tenderers can receive up to 3,000,000 SEK in price reduction for demonstrating high emission-free coverage of daily production and up to 2,000,000 SEK for emission-free transport logistics, a combined maximum of 5,000,000 SEK of the 8,000,000 SEK total possible price adjustment attributable to electrification performance. Slakthusområdet E02's instrument design is documented differently: the award criterion is not price but lowest CO2e in a climate calculation submitted via Anavitor (AFB.53), within a fixed budget of 70 MSEK. The contract compensation model (AFB.14, Entreprenadkontrakt §6.1-6.2) is running account against target cost with open books, meaning the contractor's cost premium for electrification is reimbursed rather than built into a fixed price, a structural inversion of the risk logic in the other municipal cases. The documents therefore support the interview-based finding that procurement instruments do not simply differ in how they promote electrification, but in how they allocate cost uncertainty, performance obligations and verification burdens between procurer and contractor.

## 5.3 Conditions for participation across the value chain

The empirical material shows that the conditions for participation in electrified construction procurement are not determined only by whether electrification is formulated as a requirement, bonus or penalty. Instead, participation conditions are shaped by how feasible the procurement design appears to actors who need to plan, finance, coordinate and deliver the electrified solution. This means that participation is affected by several connected factors: the ability to carry additional investment costs, the credibility of long-term demand, the maturity of the electric-machine market, access to charging infrastructure, and the ability to coordinate requirements through the supply chain.

A recurring theme in the interviews is that electrification may affect actors differently depending on their size and position in the supply chain. In the Slakthusområdet case, C1 reflects that some actors may struggle to follow the electrification transition and risk being

excluded from future work if they cannot adapt. However, P1 offers a more nuanced interpretation and argues that electrification does not necessarily exclude smaller actors by itself. Instead, he emphasises that exclusion is more likely to stem from poorly designed procurement and unfair risk allocation. From this perspective, a smaller actor should in principle be able to invest in one electric machine for a smaller project in the same way that a larger actor might invest in several machines for a larger portfolio, provided that the contract gives reasonable conditions for carrying that investment. The issue is therefore not only actor size, but whether the procurement design creates a realistic relationship between investment, risk and expected return.

At the same time, the contractor perspective indicates that larger organisations may have certain advantages in the transition. C1 describes how Skanska has a specialised division focused on environmental innovation, which makes the transition towards electrification not only a cost or risk, but also a possible source of strategic positioning. Larger contractors may therefore be able to use internal competence, financial capacity and project portfolios to turn electrification into a competitive advantage. This does not mean that smaller actors are necessarily excluded, but it indicates that the same procurement requirement can be interpreted differently depending on organisational capacity. For actors with fewer resources or a narrower project base, the same investment may appear more constraining.

A central condition for participation is the ability to interpret electrification as a durable market direction rather than as an isolated project-specific demand. Several interviewees connect investment feasibility to long-term demand signals. P2 describes a core mismatch between the length of individual construction contracts and the economic life of electric machinery. If an electric machine needs to be written off over several years, a single project may not provide enough certainty to justify the investment. From this perspective, the value of Trafikverket's bonus model is not only the individual payment, but the signal that electrification will be rewarded across future projects. If contractors believe that they can receive a bonus repeatedly, the investment becomes easier to justify over a longer period.

> *"Regardless of which contract I win with the Swedish Transport Administration, the bonus is in place during this period." – P2*

Similar reasoning appears in the municipal cases, although through other instruments. P1 describes how broader framework agreements and repeated climate requirements can create a more stable demand signal than isolated projects. P4 similarly emphasises the importance of consistency over time, where repeated requirements from larger procuring organisations can help move the market by making future demand more predictable. C3 adds the supplier-side perspective by linking feasibility to contract duration and leasing conditions. When the duration and structure of the contract allow contractors to match the use of electric equipment with financing or leasing arrangements, electrification becomes easier to price and plan. Another aspect connected to this was brought up by the contractor at Östra Fyrislund, the need for a red thread across procurements, in line with the previous findings although highlighting the technical specification of what machines are requested. This is a distinct reflection in the empirical material, but it stands to reason that investment in electric

machines require that specific machine to be requested for future projects not just an electric machine in general.

> *"In the previous procurement they put out, they had a bonus for an electric wheel loader […] But then in the next procurement they ignore the electric wheel loader entirely and ask for an electric excavator instead. That makes it difficult to invest." – C4*

The procurement documents for Slakthusområdet E02 offer a contrasting example. Document 01.02 specifies electric machines across several categories simultaneously: at least two excavators equivalent to more than 24 tonnes diesel-driven, at least one wheel loader equivalent to more than 20 tonnes, and at least two tipper trucks with a minimum load capacity of 28 tonnes, all to be met within six months of contract signing. This multi-category specification sends an investment signal across machine types at once rather than directing it toward a single category. Whether such breadth is preferable to depth in any given procurement depends on market conditions, but the contrast with C4's description of single-machine specifications switching between procurements illustrates how specification design shapes the investment logic facing contractors.

Ultimately when the time horizon is too short, uncertainty is instead priced into the offer or transferred further down the supply chain. This suggests that contract architecture, including duration, options and repeated procurement signals, functions as a condition for participation in its own right.

The empirical material also shows that participation conditions change as the market matures. Electrified construction machinery is not described as a fixed technological category, but as a rapidly developing market. P1 describes how Stockholm moved from earlier projects with lower electrification levels toward higher requirements in later projects, including the increase from 25 % to 50 % electric operation between related project stages. P4 similarly describes a planned increase in electrification requirements over time, reflecting an expectation that the market will be more capable in future procurements. P3's reasoning around Mäster Samuelsgatan also follows this logic: the project was treated as a way to learn and test what could be achieved, with the expectation that future procurements could use more direct requirements once knowledge had improved.

> *"We raised the requirement to at least 50 percent electric operation. That was something we had already set as a requirement from the start at 25 percent electric operation." – P1*

This creates a moving boundary between ambition and feasibility. C3 describes the market in 2024 as still affected by limited supply and long lead times for electric machines, while C1 contrasts earlier projects where battery capacity and charging needs created significant planning challenges with later situations where electrified operation had become more routine. P4 also highlights how quickly electric-machine availability changes, where machines available one year may differ substantially from those available the next. This

means that procurement requirements can become outdated quickly if they are not adjusted to market development. It also explains why several procurers describe a gradual escalation of requirements rather than an immediate shift to full electrification across all projects. The requirement level is not only a climate target, but also a judgement about what the market can currently deliver.

Charging infrastructure appears as another central enabling condition for participation and implementation feasibility. Across the interviews, access to electricity is described not as a secondary logistical issue, but as one of the conditions that determines whether electrification can be delivered in practice. The cases show different approaches to this responsibility. In the municipal cases, the procuring organisation appears to take a more active role in preparing or mapping electricity access. P4 describes the importance of identifying locations with sufficient grid capacity before procurement, while P3 refers to work with the grid operator through the STOLT collaboration to understand available connections in Stockholm. C3 also emphasises that contractors depend on the procurer having investigated the charging conditions before the contract starts, since this affects what machines and transport solutions can realistically be used. Similarly, C4 describes positively how the client had prepared a charging point which they could easily transfer the contract for and how that should be a part of the materials for tendering since the grid access and capacity is crucial.

> *"I absolutely think the party carrying out the procurement should have a handle on that beforehand, partly because the capacity of the electricity grid is limited in several areas." – C4*

Trafikverket's role is described differently. P2 explains that Trafikverket does not supply electricity in the same way that it would not normally supply diesel or HVO to contractors. However, Trafikverket has included discussions with grid providers before projects in order to provide bidders with information about electricity access points and available capacity. This does not remove the contractor's responsibility for planning its own production setup, but it reduces uncertainty by making the conditions clearer before tendering. In this way, charging information becomes part of the procurement's feasibility conditions even when the procurer does not take direct responsibility for providing electricity.

The interviews also show that charging infrastructure can influence the practical organisation of the construction site. P1 notes that the placement and design of charging solutions may affect which machines are suitable for a project. For example, a central charging area may favour some machine types and work patterns over others. In other situations, it may be more efficient for the contractor or subcontractor to arrange charging as part of its own production planning, especially where the relevant competence already exists within the supplier chain. This indicates that the question is not simply whether the procurer or contractor "owns" the charging responsibility. Rather, the important point is that responsibility for charging needs to be clear enough for contractors to plan, price and coordinate electrified operation.

> *"At least 50 percent electric operation [...] that meant we placed even greater emphasis on planning the infrastructure." – C1*

The supply-chain dimension adds another layer of complexity. Electrification requirements are often placed in the contract between procurer and main contractor, but much of the actual machine and transport operation may be performed by subcontractors or suppliers. This means that the requirement must be translated through several contractual and operational layers before it becomes actual electric operation on site. The interviews suggest that this translation is not automatic. P3 reasons that a 100 % requirement at the main-contractor level should flow down to the subcontractors, since the head contractor must ensure that all actors used in production meet the project requirements.

In practice, however, other interviews show that this cascade depends on more than contractual wording. P1 points to the importance of OEM investment, since subcontractors and contractors can only deliver electrification if suitable machines are available in the market. C3 describes situations where traditional suppliers had not invested in electric equipment, forcing contractors to look for alternative suppliers. P2 adds a reporting dimension through Trafikverket's use of digital systems, where energy use from different actors can be incorporated into the same reporting infrastructure. This suggests that cascading requirements through the value chain requires coordination between formal contract requirements, equipment availability, contractor procurement flexibility and follow-up systems.

> *"Traffic arrangements, which end you start from, production planning, temporary structures [...] when it starts to become a bit more complex, it is usually worth doing." – C1*

Overall, the findings show that participation conditions are shaped by the interaction between procurement design and the practical capacity of the supply chain. A requirement may be clear on paper but difficult to deliver if the market lacks available machines, if charging conditions are unclear, if the contract horizon is too short, or if risks are pushed onto actors without the capacity to carry them. Conversely, electrification becomes more feasible when procurers provide predictable demand signals, clarify enabling conditions, and design contracts that allow contractors and subcontractors to plan investments over time. In this sense, participation is not only a question of whether actors are formally allowed to tender, but whether the procurement design gives them workable conditions for doing so.

## 5.4 Verification and follow-up of electrification requirements

Verification is central to whether climate-oriented procurement instruments become credible in practice. If electrification requirements, bonuses or penalties cannot be followed up, their practical effect becomes uncertain. Across the interviews, electrification stands out as a climate requirement that is comparatively easy to verify in some respects, but that also requires more detailed data infrastructure when the instrument depends on measured energy use or quantified climate performance.

The follow-up structures embedded in the procurement documents illustrate how verification logic varies by instrument type. Östra Fyrislund's administrative requirements mandate monthly reporting of all vehicles, machines, fuels and hydraulic fluids via a standardised Excel template (document 13.20, AFC.2231), with the procurer reserving the right to conduct spot-checks (AFC.2231). The contractor is required to conduct an internal environmental and quality audit three months after construction starts, with the procurer entitled to attend (AFC.2252). External inspection costs are charged to the contractor if deviations are found at audit (AFC.115). Trafikverket's verification system is more technically demanding: bonus payment requires digital delivery notes submitted via BEAst Supply standard over Peppol, with no schematic calculations permitted, only measured actual consumption is accepted. Mäster Samuelsgatan's contract (AFC.13 and AFC.242) requires the contractor to use a digital follow-up tool reporting energy consumption and CO2 savings per machine type and transport, with monthly submissions no later than the fifth working day of each month and a final summary report on energy use and emissions reduction at project close. These formal structures confirm the layered verification picture that emerges from the interviews: basic electrification presence is observable, but bonus-based and performance-based instruments require data infrastructure that goes substantially beyond visual site inspection.

Electrification has a lower verification threshold than many other environmental requirements because compliance is directly observable. P1 describes electric machines as relatively simple to verify because the procurer can inspect the site and see whether a machine has an exhaust pipe or not.

> *"You go out and take a look; does it have an exhaust pipe? No, good. When the requirements are as clear-cut as electric machines, it is easy." – P1*

This makes electrification different from requirements related to fuels, oils or calculated climate impact, where compliance may require sampling, documentation or agreed calculation methods. P1 contrasts this with requirements concerning biodiesel or other fuels, where the procurer may need external support or chemical testing to determine what fuel is actually being used.

> *"Requirements for biodiesel, certain types of oils, in those cases we have a consultant who goes out and takes chemical samples. What is actually in the tank?" – P1*

He also describes carbon calculations as more methodologically dependent, since the parties first need to agree on how emissions are calculated and which tool should be used.

> *"When you talk about CO2 impact, you have to agree on how the calculations are to be done. We chose to use Anavitor as the basis." – P1*

A related distinction concerns how the requirement is formulated. P4 makes a related point from a different angle. In his view, requirements expressed in fixed quantities, percentages or machine use are easier to follow up than more complex performance-based climate

calculations. P3 also notes that a full electrification requirement can be easier to verify than partial requirements, since a 100 percent requirement reduces the need to draw boundaries around what counts and what does not. A partial requirement, such as a certain share of machine hours or energy use, creates a need to measure the boundary between compliant and non-compliant operation. This means that the ease of verification depends not only on whether the object is electric, but also on how the requirement is formulated.

However, visual inspection is not sufficient for all types of electrification instruments. The Trafikverket bonus model shows a different verification logic, since payment depends on the amount of electricity used. In this case, the relevant question is not simply whether an electric machine or vehicle is present, but how much electricity is actually used in eligible work. P2 describes Trafikverket's reporting system as based on digital follow-up of fuel and energy use, where the data can be checked against the requirements of the bonus model. This makes the bonus model more dependent on data quality and reporting infrastructure than a simple machine requirement.

> *"It has to be actual consumption […] you cannot sit in the office and say, it has probably consumed 48 kilowatt-hours today." – P2*

This distinction shows that different procurement instruments require different verification systems. Requirement-based instruments can often be verified through a combination of site inspection and periodic reporting, especially when the requirement concerns the presence or use of specific electric machines. Bonus-based instruments require more precise measurement because payment depends on actual performance. Penalty-based instruments require sufficient evidence to determine whether a deviation has occurred and whether a sanction should be applied. Verification is therefore not only an administrative routine after contract award, but part of the procurement instrument's design logic.

The interviews also show that follow-up can involve several layers. P1 describes cases where Stockholm Stad has used external or independent certification in connection with infrastructure and climate requirements. P4 describes a follow-up approach where sustainability coordination on site becomes important for checking whether requirements are met. C1 adds the contractor perspective by describing how reporting and aggregation may in practice involve spreadsheets and internal data collection. This indicates that even when the requirement itself appears simple, the follow-up process can create administrative work and potential sources of error. Data needs to be collected, entered, aggregated and reported in a way that both the contractor and procurer can rely on.

> *"It becomes a lot of data to gather from various places […] the problem with Excel is sometimes when it goes wrong, it can go very wrong." – C1*

The contractor perspective is important here because follow-up is not only something imposed by the procurer; it becomes part of the contractor's production management. Where several subcontractors, machines and transport flows are involved, the main contractor needs

to collect information from several actors and translate it into the format required by the client. This is especially relevant when requirements apply to both machines and transports, or when the procurer wants information about energy use and emissions reduction rather than only confirmation that a machine is electric. In such cases, verification becomes linked to the coordination problems discussed in Section 5.3.

The empirical material therefore suggests that electrification has a dual character in relation to verification. On one hand, it is comparatively visible and therefore easier to verify than many fuel-based or calculation-based climate requirements. On the other hand, more advanced electrification instruments, such as bonuses based on kilowatt-hours or requirements expressed as shares of energy use, require reliable data systems and clear reporting routines. The simplicity of verifying “electric or not electric” does not remove the need for more detailed follow-up when the procurement instrument is tied to measured performance.

This has implications for the credibility of different procurement designs. A strict requirement is only credible if the procurer can identify non-compliance. A bonus is only credible if the contractor can demonstrate eligible performance. A penalty is only legitimate if deviations can be documented clearly enough to justify the sanction. Verification therefore shapes both enforceability and perceived fairness. If the follow-up system is unclear or burdensome, it may increase uncertainty for contractors. If it is clear and proportionate, it can reduce ambiguity and make the procurement instrument easier to accept.

Overall, verification and follow-up are not secondary technical details, but part of how procurement instruments function in practice. Electrification offers an advantage because electric machines and vehicles can often be observed more directly than other climate measures. However, the cases also show that the verification burden depends on the level of precision required by the instrument. The more the procurement design moves from simple requirements toward quantified bonuses, energy shares or calculated climate outcomes, the more important digital reporting and agreed verification routines become. This makes verification a central condition for translating climate ambition into enforceable procurement practice.

# 6. Discussion

The purpose of this thesis was to examine how procurement instrument designs in electrification-oriented Swedish construction procurements structure risk allocation and responsibility positioning. Additionally, it was to examine what procurer and contractor perspectives reveal about participation conditions, implementation feasibility and tensions between climate ambition and practical delivery. The discussion returns to this purpose by interpreting the findings in relation to the research questions, the conceptual framework and the literature reviewed in Chapter 2.

The findings show that procurement instruments should not be understood only as administrative tools for specifying environmental ambition. In the selected cases, requirements, bonuses, penalties, competitive dialogue and follow-up routines shaped how different actors interpreted the feasibility of electrification. These instruments influenced who carried cost uncertainty, who was exposed to non-compliance, who had to coordinate enabling conditions such as charging infrastructure, and who had the practical ability to participate in the procurement. In this sense, procurement design was not a neutral container for climate objectives, but rather a mechanism through which uncertainties related to climate ambition were distributed across actors.

The discussion is structured around the three sub-questions, drawing together the thematic findings of Chapter 5 under each question. Section 6.1 addresses the procurement instrument designs in the selected cases and how they positioned risks and responsibilities. Section 6.2 discusses how procurers and contractors interpreted participation conditions and implementation feasibility. Section 6.3 discusses the main tensions that emerged between climate ambition and feasibility. Together, these sections provide the basis for answering the main research question: how procurement instrument design shapes the balance between risk allocation, actor participation and climate ambition in electrification-oriented construction procurement.

## 6.1 Instrument design and the positioning of risk (Sub-Q1)

The first sub-question asked what procurement instrument designs were employed in the selected cases and how these designs positioned electrification-related risks and responsibilities. The findings show that the four cases used different instrument logics. Slakthusområdet E02 used a requirement-based design combined with competitive dialogue and a compensation model that reduced the contractor's cost uncertainty. Trafikverket's bonus model used a voluntary incentive structure, where the contractor decides whether to electrify and receives compensation if measured electricity use is reported. Mäster Samuelsgatan/Jakobsbergsgatan combined competitive dialogue, qualitative evaluation and penalties to support a high electrification ambition. Östra Fyrislund used a requirement-based design with a fixed high penalty attached to the electric-excavator requirement.

The comparison shows that instrument design positioned risks differently. Bonus-based instruments, as seen in Trafikverket's model, did not force electrification in each project. Instead, they left the operation decisions to the contractor while allowing the procurer to absorb part of the investment risk through financial compensation. This aligns with the literature on incentives as a way of influencing supplier behaviour without necessarily excluding actors at the tendering stage. At the same time, the bonus model also illustrates the limits of voluntary instruments: if the bonus is too low or too uncertain, it may not be sufficient to change investment behaviour. The findings therefore nuance the idea that incentives are inherently less risky. They may reduce participation risk, but they can also produce weaker climate steering if they are not sufficiently calibrated. This can be related to performance-based contracting literature, where incentives are used to align supplier behaviour with buyer objectives through performance-linked payment mechanisms

(Molitoris, Glas and Eßig, 2025). However, Trafikverket's model differs from a stricter performance-based contract because electrification remains voluntary: the contractor is not punished for not electrifying, but is compensated if it chooses to do so and can verify eligible electricity use. Overall, the findings indicate that bonuses may have a role when combined with a base requirement, for example as an incentive for going beyond the required level of electrification.

Requirement-based instruments positioned risk differently. In Slakthusområdet and Östra Fyrislund, electrification was formulated as something that had to be delivered rather than something that could be voluntarily rewarded. This created a clearer climate ambition and reduced ambiguity over the expected outcome. However, it also increased the importance of feasibility assessment before contracts are awarded. If the market cannot supply the required equipment, if charging infrastructure is unclear, or if the contractor must carry cost uncertainty under a fixed-price model, the requirement can become a source of increased risk pricing or reduced participation, lowering competition. This supports the argument by Kadefors *et al.* (2021) that strict requirements may reduce competition if not carefully designed, but the cases also show that requirements do not have a uniform effect. Their implication depend on the surrounding contract design. The cases therefore support Kadefors *et al*.'s (2021) concern about exclusionary effects, but refine it by showing that strictness alone is not the decisive factor. A requirement becomes more or less exclusionary depending on whether it is paired with dialogue, preparation time, cost compensation, charging clarity and feasible verification.

The Slakthusområdet case is especially important since it shows that a requirement-based instrument does not automatically mean that the contractor carries all electrification-related risk. The formal contract structure separated cost uncertainty from the climate-performance responsibility. The municipality carried much of the cost uncertainty through the running-account model, while the contractor remained responsible for achieving the agreed climate-performance ceiling. This complicates a simple requirement-versus-bonus distinction. The same instrument type can allocate risk differently depending on compensation model, dialogue process and follow-up structure. This relates to the literature by Li, Arditi and Wang (2015) and how they describe transaction cost as closely linked to contract formulation and the risk distribution therein. This connects directly to Li, Arditi and Wang's (2015) argument that risks allocated to a party poorly positioned to manage them tend to reappear as contingencies, higher mark-ups or disputes. Slakthusområdet illustrates the inverse logic: by separating cost uncertainty from climate-performance responsibility, the contract reduced the need for the contractor to price all electrification uncertainty into the tender while still retaining accountability for the agreed emissions ceiling. The case therefore shows how proportionate risk allocation can support both climate ambition and contractual acceptance.

Penalty-based elements also had dual implications. On one hand, penalties strengthened enforceability and protected the credibility of electrification requirements. The Mäster Samuelsgatan contractor's support for strong penalties indicates that penalties can be viewed as a mechanism for fair competition, since they prevent bidders from promising electrification without intending to deliver it. On the other hand, the Östra Fyrislund

contractor's experience suggest that penalties also can become a pricing uncertainty when requirements are difficult to interpret or when compliance risk is affected by changing project conditions. This duality reflects two strands of the literature. From a contracting perspective, penalties can discipline performance but may also increase contingencies when the risk is difficult to price or control (Li, Arditi and Wang, 2015). From a GPP perspective, penalties are only meaningful if environmental requirements are monitored and followed up, since weak follow-up can make sustainability criteria symbolic rather than operational (Varnäs, Balfors and Faith-Ell, 2009; Cheng *et al.*, 2018).

Competitive dialogue emerged as a cross-cutting mechanism that modified how the other instruments functioned. Where it was used (Slakthusområdet, Mäster Samuelsgatan/Jakobsbergsgatan), it allowed procurers to calibrate requirements against actual market capacity before locking in contract terms, reducing the risk that requirements would be either too loose or undeliverable. Where it was absent (Östra Fyrislund), the same knowledge gap had to be absorbed through preparation time, procurer expertise and, from the perspective of C4, higher risk-pricing in tenders. Crucially both P1 and P3 describes dialogue as a temporary tool: once institutional knowledge matures the competitive dialogue becomes redundant and procurers migrate toward traditional procurement processes. This positions competitive dialogue not as a permanent feature of good procurement design but as a learning mechanism, advantageous in procurements with added complexity. This extends the perspective on competitive dialogue expressed by Sönnichsen and Clement (2020) from a relation building opportunity to include inherent knowledge building opportunities for the actors involved. This also relates to broader SPP/GPP literature identifying lack of knowledge, market information and organisational capability as recurring barriers to sustainable procurement implementation (Brammer and Walker, 2011; Cheng *et al.*, 2018). In the cases studied here, competitive dialogue functioned as one way of addressing this barrier by importing market knowledge before the requirements were finalised.

Overall, the findings suggest that electrification-related risks need to be allocated to the actors best positioned to manage them, rather than simply transferred to the contractor. When risk is poorly allocated, the findings indicate that it may return as higher prices, reduced participation or conflict between parties. Additionally, procurements focused purely on cost minimization, although a staple in public procurement regimes, risks being sub-optimal in situations with increased uncertainties, something both procurers and contractors reflected upon in the findings.

The first sub-question can therefore be answered as follows: the selected cases employed requirement-based, bonus-based, penalty-centred and dialogue-supported procurement designs, and these designs positioned risks differently across the procurer–contractor relationship. Bonuses mainly addressed investment risk by compensating voluntary electrification. Requirements created clearer performance obligations but increased the importance of feasible contract conditions. Penalties strengthened enforcement but could increase risk pricing if the underlying requirement was ambiguous. Competitive dialogue and compensation models modified these effects by allowing some risks to be clarified, shared or

reallocated before contract execution. Instrument choice should therefore be understood as a risk-positioning decision rather than only as choice between alternative climate-policy tools.

## 6.2 Participation conditions and implementation feasibility (Sub-Q2)

The second sub-question asked how procurers and contractors interpreted the implications of different procurement instrument designs for participation conditions and implementation feasibility. The findings show that procurers and contractors often interpreted feasibility from different positions in the procurement system.

For procurers, feasibility was largely connected to the possibility of moving the market without creating unrealistic requirements. In the cases with competitive dialogue, procurers used the procedure to understand what contractors, OEMs and subcontractors could deliver before finalising the contract. In cases without competitive dialogue, feasibility depended more heavily on the procurer's prior knowledge, preparation time and ability to define requirements clearly. This indicates that procurers interpret feasibility partly as a knowledge problem: what can the market currently deliver, and how should this be translated into procurement documents? This corresponds to GPP implementation literature, where limited environmental knowledge and insufficient integration with industry actors are recurring barriers to translating sustainability ambition into procurement practice (Sourani and Sohail, 2011; Cheng *et al.*, 2018). The findings indicate that this can be managed in different ways but that the use of competitive dialogue is uniquely well suited to improve the knowledge asymmetry.

For contractors, feasibility was more directly connected to production planning, equipment access, charging logistics, cost compensation and risk exposure. Contractors did not interpret electrification only as a question of whether electric machines exist. They assessed whether those machines could be obtained in time, whether they could perform the work efficiently, whether charging could be organised without disrupting production, and whether the contract allowed the additional cost and uncertainty to be recovered. These concerns mirror electrification-adoption literature, which identifies high upfront costs, charging requirements, battery/range limitations and productivity effects as central barriers for electric construction machinery (Stokke *et al.*, 2023; Yi *et al.*, 2024, 2026). Additionally, this supports the conceptual framework's distinction between formal participation and practical implementation feasibility. A contractor may be formally eligible to tender while still perceiving the procurement as risky, costly or operationally difficult.

The findings also show that participation conditions depend on the durability and coherence of demand signals. This relates directly to Lundberg, Marklund and Strömbäck's (2016) discussion of green public procurement as signalling mechanism. Contractors and suppliers need to believe that electrification demand will persist beyond one isolated project in order to justify investment. However, the findings add a more specific point: demand signals must be technically coherent. It is not enough for procurer to repeatedly request "electric machinery" in general. If one procurement asks for an electric wheel loader and the next asks for an

electric excavator, the signal may still be too fragmented for contractors or suppliers to justify a specific equipment investment. This refines Lundberg, Marklund and Strömbäck's (2016) signalling argument. Their point is that repeated public procurement demand can allow suppliers to distribute investment costs over time. The present findings suggest that, in electrified construction, this signal must also be technically specific enough to support investment in particular machine categories. Participation conditions are therefore shaped not only by the presence of future demand, but by whether that demand is predictable enough to support investment planning.

Actor size and supply-chain position also shaped feasibility. Larger contractors appeared better positioned to absorb uncertainty, mobilise internal competence and treat electrification as a strategic opportunity. Smaller actors or lower-tier suppliers may experience the same requirements as more constraining, particularly when they cannot spread investment costs across a broader portfolio. This interpretation is consistent with construction supply-chain literature describing construction as fragmented and dependent on subcontracted production (Segerstedt and Olofsson, 2010), as well as research showing that cost-driven subcontractor selection can complicate innovation and coordination across firms (Hartmann and Caerteling, 2010; Havenvid, Håkansson and Linné, 2016). It also connects to McDermott and Hayes' (2018) argument that risk can shift down multi-tier contracting chains to actors less able to manage it. However, because lower-tier actors were not directly interviewed, this thesis treats cascading risk as an analytically indicated concern rather than a fully demonstrated empirical outcome.

Charging infrastructure emerged as a central enabling condition. The findings show that charging is not only a technical issue but a procurement-design issue. This supports Stokke *et al.*'s (2023) finding that energy supply and charging infrastructure are central conditions for zero-emission construction sites, and Yi *et al.*'s (2024; 2026) emphasis on charging time, range and productivity as barriers to electric construction machinery. The contribution of the present study is to show that these are not only technical barriers, but procurement-design variables: uncertainty about charging determines how risk is priced and whether electrification appears feasible. Where the procurer prepares electricity access or provides reliable grid information, implementation uncertainty is reduced. Where responsibility for charging is unclear, or where the project starts before grid capacity exists, electrification becomes more difficult regardless of the formal procurement instrument used. This means that procurement requirements for electric machinery can only be assessed once a proper understanding of the charging infrastructure situation is attained.

The second sub-question can therefore be answered as follows: procurers and contractors interpret participation and feasibility differently but in interdependent ways. Procurers focus on whether the procurement design can steer the market while remaining economically defensible. Contractors focus on whether the design provides workable conditions for planning, pricing, sourcing equipment and coordinating production. Participation is therefore not only determined by formal eligibility criteria. It is shaped by whether the procurement design creates credible, consistent and practically manageable conditions for actors across the value chain. As such participation conditions cannot be specified by procurers in isolation;

they emerge through the alignment of procurer ambition with contractor perspectives on operational reality. Dialogue is one mechanism through which this confrontation can be made explicit, although the cases also show that prior procurer knowledge and preparation time can partly serve a similar function.

## 6.3 Tensions between climate ambition and implementation feasibility (Sub-Q3)

The third sub-question asked what tensions emerge between climate ambition and implementation feasibility across the selected cases. The findings show that these tensions are not accidental side effects. They are central to electrification-oriented procurement because the procurer is trying to use a contractual process to accelerate a technological and organisational transition that is still developing.

The first tension concerns ambition and participation. Requirements can create stronger climate steering than voluntary instruments, but they may also narrow participation if they are introduced before the market has sufficient capacity. This tension is visible in the contrast between Trafikverket's bonus model and the municipal requirement-based cases. Trafikverket's approach avoids excluding contractors who cannot yet electrify, but it may not produce rapid transition unless the bonus is strong and predictable enough. Requirement-based cases create clearer transition pressure, but they also place greater importance on preparation, dialogue and risk sharing. This tension reflects the dilemma identified in GPP literature between ambitious environmental criteria and competition/cost efficiency (Cheng *et al.*, 2018). It also echoes Kadefors *et al.*'s (2021) finding that strict environmental requirements can become exclusionary, while less mandatory criteria may preserve participation but provide weaker steering. As such, this suggests that there is no universally superior instrument. The appropriate design depends on the maturity of the market, the procurer's knowledge and the feasibility of the specific project.

The second tension concerns consistency and project specificity. Contractors need long-term, coherent demand signals in order to justify investment in electric machinery. At the same time, the cases show that electrification feasibility is highly project-specific. Inner-city projects may be suitable due to shorter transport distances, greater charging access and stronger local climate-policy pressure. Other projects may face more difficult grid conditions, longer transport distances or machine categories that are not yet mature. A general rule requiring electrification across all projects could therefore accelerate market transition, but it could also force electrification into contexts where the cost and complexity become disproportionate. C4's reflections on projects where the electricity grid is not yet available indicate that universal electrification requirements may become disproportionate unless enabling infrastructure is addressed first. In such cases, interviewees suggested that other lower-emission measures, such as HVO, may be more feasible than full electrification. Conversely, leaving too much discretion to individual procurers may preserve project-specific judgement, but can also make electrification dependent on motivated individuals and local initiative making the transition rate a function of personnel rather than policy, and therefore structurally fragile as a decarbonisation mechanism. This tension links back to the findings on

mandate, individual championship and institutional learning. The cases therefore indicate that a middle path may be preferable: escalating baseline requirements over time, while still allowing project-level feasibility considerations to shape the specific instrument design.

The third tension concerns enforcement and risk pricing. Penalties are important because they make requirements credible. Without follow-up and sanctions, sustainability requirements may risk becoming symbolic, a problem identified in GPP literature on weak monitoring and evaluation (Varnäs, Balfors and Faith-Ell, 2009; Cheng *et al.*, 2018). However, penalties can also increase uncertainty if the underlying requirement is unclear or if compliance depends on conditions outside the contractor's control. In those situations, the penalty may be priced into the tender or may discourage participation, which reflects Li, Arditi and Wang's (2015) argument that difficult-to-control risks may return as contingencies or higher prices. The findings therefore suggest that penalties are most useful when they are attached to requirements that are clear, measurable and feasible.

The fourth tension concerns verification and measurement. Electrification has an advantage compared with several other climate criteria because electric machines are comparatively easy to observe. In this sense, electrification may reduce some of the measurability problems identified in earlier research on green procurement. However, the findings also show that visibility is not the same as complete verification. Bonus-based models and requirements based on energy shares require reliable data systems, actual consumption reporting and clear routines for aggregation across machines, subcontractors and transport flows. Electrification is therefore both easier and more demanding to verify: easier when the question is whether a machine is electric, but more demanding when payment, penalty or performance evaluation depends on measured use. This finding speaks directly to the literature on measurability and follow-up in green procurement. Varnäs, Balfors and Faith-Ell (2009) and Cheng *et al.* (2018) identify weak monitoring and evaluation as recurring barriers in GPP implementation, while Kadefors *et al.* (2021) show that climate calculations and reporting can increase contractors' tendering and administrative burden. The present findings suggest that electrification partly reduces this problem when compliance is physically visible, but reintroduces it when instruments rely on measured kWh, electrification shares or aggregated emissions data.

Taken together, the third sub-question can be answered by arguing that the main tension is not simply between high climate ambition and contractor resistance. Rather, the tension lies between climate ambition and feasible risk absorption. Public procurers can raise ambition through stricter requirements, bonuses, penalties and follow-up, but each instrument creates a different distribution of cost uncertainty, compliance exposure, decision responsibility and administrative burden. The role of procurement design is therefore not to remove tension entirely, but to make it manageable by aligning ambition with market maturity, enabling conditions and actor capacity.

## 6.4 Relating findings to the conceptual framework

The conceptual framework developed in Chapter 3 positioned procurement instrument design as the starting point for analysing how electrification-oriented procurement shapes risk allocation, responsibility positioning, participation conditions and implementation feasibility.

The findings support this basic logic. Across the four cases, the formal procurement instrument was not merely a technical or administrative choice. Instead, it structured what kind of uncertainty actors were exposed to, who was expected to manage that uncertainty, and under what conditions contractors could interpret electrification as practically feasible.

The framework was useful because it made it possible to compare instruments that at first appear difficult to compare. Trafikverket's bonus model, Slakthusområdet's requirement-based model with running-account compensation, Mäster Samuelsgatan's penalty-centred competitive dialogue process and Östra Fyrislund's fixed-penalty requirement all aimed to support electrification. However, they did so through different combinations of requirement, incentive, sanction and verification. By analysing these instruments through the framework, the cases could be compared not only according to their level of climate ambition, but according to how they positioned investment risk, compliance risk, enabling-condition risk and verification burden.

The four risk categories proposed in Section 3.2 also held analytical value across the cases, although with different degrees of prominence. Utilisation risk surfaced most clearly in contractor reflections on bonus- and requirement-based instruments, where the question of whether electric equipment would be used enough to justify acquisition, leasing or deployment shaped pricing and investment willingness. P2's framing of the Trafikverket bonus as available across future contracts during the relevant time period can be read as a response to this utilisation uncertainty, since it allows contractors to view electrification investment across more than one project. Financing and CAPEX risk was most visible in Östra Fyrislund, where the fixed-price model positioned cost-premium exposure primarily with the contractor, and in C4's reflection on the difficulty of justifying investment when requested machine categories change between procurements. Compliance and penalty risk was central to both Östra Fyrislund and Mäster Samuelsgatan, although with different contractor interpretations: C4 described penalties as pricing uncertainty during tendering, while C3 supported penalties as protection against opportunistic bidding. Enabling-condition risk, particularly relating to charging infrastructure and grid capacity, emerged as especially central in the empirical material and appeared more prominent than initially expected in the framework.

At the same time, the findings refine the framework by showing that instrument type alone is not sufficient to explain risk allocation. A requirement-based procurement does not automatically mean that the contractor carries all relevant risk. Slakthusområdet shows that a strict climate requirement can be combined with a compensation model where the procurer absorbs much of the cost uncertainty, while the contractor remains accountable for climate-performance outcomes. Östra Fyrislund, by contrast, shows how a requirement-based model with fixed price and high penalties can expose the contractor more directly to cost and compliance uncertainty. The same broad category of instrument can therefore produce different risk distributions depending on the surrounding contract architecture.

A similar point applies to bonus-based instruments. In the conceptual framework, bonuses were treated as instruments that can reduce exclusion risk by allowing voluntary uptake

rather than imposing mandatory requirements. The findings support this, particularly in the Trafikverket case. However, they also show that bonuses only become meaningful if they are calibrated against the actual cost and uncertainty faced by contractors. A bonus that is too low may preserve participation but fail to generate behavioural change. A bonus that is predictable over time may instead contribute to investment feasibility by signalling continued demand. The framework is therefore strengthened by distinguishing between the formal presence of an incentive and the practical strength of that incentive in relation to contractor decision-making.

The findings also show that risk allocation is strongly shaped by cross-cutting mechanisms. Competitive dialogue was treated in the framework as a pre-contract mechanism that can reduce information asymmetry and clarify feasibility. The findings support this although they also suggest that dialogue has a broader role as an institutional learning mechanism. In both Slakthusområdet and Mäster Samuelsgatan, competitive dialogue allowed procurers to access market knowledge before requirements were finalised. Over time, this knowledge could become internalised within the procuring organisation, making competitive dialogue less necessary in later procurements. This indicates that dialogue does not only affect one procurement. It can also build organisational capability for future procurement design.

Verification and follow-up also became more central than expected. In the framework, verification was understood as a post-contract mechanism that affects enforceability. The findings confirm this but also show that verification influences perceived fairness and feasibility. A penalty-backed requirement is only legitimate if non-compliance can be identified clearly. A bonus is only credible if eligible performance can be measured. A requirement is only meaningful if the procurer has the ability to follow up whether it is being met. Verification therefore does not simply occur after contract award; it is part of the design logic of the instrument itself. This is especially important in electrification procurement because different instruments require different forms of verification, ranging from visual inspection of machines to digital reporting of actual electricity use.

The framework's emphasis on actor size and supply-chain position also remains relevant, although the empirical material requires caution. The findings suggest that larger contractors may be better positioned to absorb uncertainty, leverage internal expertise and treat electrification as a strategic opportunity. They also suggest that risks can cascade through subcontracting relationships, particularly when main contractors depend on suppliers or machine owners for electric equipment. However, because lower-tier actors were not directly interviewed, the thesis cannot claim to fully explain how subcontractors or machine owners experience these requirements. The framework is therefore useful as a sensitising lens rather than as a basis for definitive claims about all supply-chain actors.

Finally, the findings clarify the relationship between participation conditions and implementation feasibility. These concepts were separated in the framework, but the empirical material shows that they are closely connected. Participation is not only a question of whether actors are formally allowed to tender. It also depends on whether the procurement design creates workable conditions for planning, pricing, sourcing equipment, arranging charging and managing compliance. A procurement can therefore be formally open while still

being practically difficult for some actors to enter. Conversely, a strict requirement can be feasible if the surrounding design reduces uncertainty through preparation time, dialogue, adequate cost compensation, charging infrastructure clarity and credible verification.

One finding emerged that extends the framework rather than merely refining it. The framework assumed procurement instrument design as the analytical starting point, but Section 5.1 shows that the decision to use electrification-oriented instruments at all rests on a prior commitment that is anchored differently across the cases. In Östra Fyrislund and Slakthusområdet, this anchoring was primarily political. In Trafikverket, it was embedded in organisational policy. In Mäster Samuelsgatan, it was driven more strongly by individual initiative. This is analytically relevant because the durability and consistency of electrification demand depends not only on instrument design, but also on whether the underlying commitment is institutionalised or personally driven. A procurement system that relies heavily on motivated individuals to initiate electrification-oriented instruments may be more structurally fragile than one anchored in mandate or organisational policy, even if the individual instruments are well designed. The framework can accommodate this finding, but it did not explicitly include anchoring as a prior condition shaping whether electrification-oriented procurement instruments are used in the first place.

Overall, the findings support the framework's central proposition that procurement design shapes electrification not directly, but through the way it structures risk, responsibility and feasibility. However, the findings also refine and extend the framework by showing that procurement instruments cannot be analysed in isolation. Their practical effects depend on the interaction between instrument type, compensation model, competitive dialogue, charging preparation, market maturity and verification routines. In addition, the decision to use such instruments at all depends on how electrification ambition is anchored within the procuring organisation. The framework therefore remains useful, but the empirical material suggests that future analysis of climate-oriented procurement should treat anchoring, dialogue and verification as integral parts of risk positioning rather than as secondary procedural details.

# 7. Conclusion

## 7.1 Answering the main research question

The main research question asked: How does the design of procurement instruments in electrification-oriented Swedish construction shape the balance between risk allocation, actor participation, and climate ambition? Drawing on four Swedish cases, Slakthusområdet E02, Trafikverket's bonus model, Mäster Samuelsgatan and Jakobsbergsgatan, and Östra Fyrislund Etapp 5-6, the thesis shows that procurement instruments shape this balance by determining how uncertainty is distributed. Requirements create clearer climate steering but increase the importance of feasibility assessment, cost compensation and enabling conditions. Bonuses reduce exclusionary pressure by preserving contractor discretion, although their climate effect depends on calibration and repeated demand signals. Penalties strengthen enforceability, but may also increase risk pricing when requirements are ambiguous or depend on conditions

outside the contractor's control. Competitive dialogue, compensation models and verification routines modify these effects by clarifying feasibility, redistributing cost uncertainty and making compliance credible. The findings indicate that the analytical unit for understanding these effects is not instrument type alone, but instrument architecture: the combination of instrument type with compensation model, dialogue, verification and enabling-condition clarity.

Across the cases, the three dimensions in the research question pulled against each other rather than aligning automatically. Higher climate ambition can narrow participation when it is translated into strict qualification criteria, mandatory requirements or penalty-backed obligations without corresponding feasibility support. This was visible in the Mäster Samuelsgatan pre-qualification process, where emission-free experience was used as a selection criterion, and in Östra Fyrislund, where the lowest-priced bidder was disqualified for failing the electric-excavator requirement. Feasibility became more workable when enabling conditions such as charging infrastructure, grid access and machine availability were addressed in procurement design rather than left as a contractor problem. Where procurement design failed to bridge ambition with enabling conditions, contractors could instead price in uncertainty or avoid participation.

These instrument-design effects were themselves conditioned by a prior factor that the conceptual framework did not anticipate. Whether electrification ambition was institutionally anchored in political mandate or organisational policy, as in Östra Fyrislund, Slakthusområdet and Trafikverket, or carried more strongly by individual initiative, as in Mäster Samuelsgatan, shaped the consistency and durability of demand. A procurement system that relies on motivated individuals to initiate electrification-oriented instruments is therefore more structurally fragile than one anchored in mandate or organisational policy, even where the individual instruments themselves are well designed.

The central conclusion is therefore that procurement instruments should be understood as risk-positioning mechanisms, not merely as alternative ways of expressing climate ambition. The balance between risk allocation, actor participation and climate ambition is not optimised by any single instrument. It is produced through trade-offs embedded in instrument architecture and conditioned by how the underlying ambition is anchored.

## 7.2 Analytical and scientific contribution

This thesis contributes to the literature on sustainable public procurement, climate-oriented construction procurement and procurement risk allocation in three main ways. In addition, the findings point to a further exploratory insight concerning the anchoring of electrification ambition.

First, the thesis shows that instrument type alone is too unrefined an analytical category to explain risk allocation in electrification-oriented procurement. The same broad instrument type can produce different risk allocations depending on its surrounding contract architecture, including compensation model, follow-up structure, market understanding and enabling-

condition clarity. This extends the contracting and risk-allocation literature by showing how risk allocation in climate-oriented procurement is shaped not only by whether requirements, bonuses or penalties are used, but by how these instruments are combined and operationalised (Li, Arditi and Wang, 2015; Molitoris, Glas and Eßig, 2025). It also complicates Kadefors *et al.*'s (2021) distinction between requirements and award criteria by showing that the implications of strict requirements depend strongly on the surrounding procurement design.

Second, the thesis extends existing conceptualisation of competitive dialogue in public procurement. Sönnichsen and Clement (2020) frame competitive dialogue as a relation-building mechanism that reduces information asymmetries between procurer and contractor before contract terms are finalised. The empirical material confirms that function, but also indicates that competitive dialogue can function as an institutional learning mechanism with implications beyond a single procurement. Both P1 and P3 described dialogue as a tool deployed while internal competence was insufficient, and both described moving away from it once competence on electrified construction procurement had matured. This situates competitive dialogue not as a permanent feature of sustainable procurement design but rather as a temporary learning device whose value may lie partly in making itself redundant.

Third, the GPP literature identifies weak monitoring as a recurring barrier (Varnäs, Balfors and Faith-Ell, 2009; Cheng *et al.*, 2018). The thesis refines this by showing that electrification does not have one fixed verification burden. Basic either/or verification, such as whether a machine is electric, can be relatively simple. However, this simplicity does not apply to all electrification instruments. Instruments based on measured energy use, electrification shares or quantified climate performance require more demanding data infrastructure and reporting routines. Verification burden is therefore not inherent to electrification as such, but depends on the instrument architecture.

In addition to these three contributions, the findings point to the importance of anchoring as a prior condition for procurement instrument use. The cases suggest that electrification-oriented procurement does not emerge from instrument design alone, but depends on whether climate ambition is anchored in political mandate, organisational policy or individual initiative. As set out in Section 6.4, this extends the conceptual framework by adding a prior condition that shapes whether electrification-oriented instruments are used at all. It is therefore presented as an exploratory extension to the framework rather than a fully established empirical contribution, and for that reason is also taken up as a direction for further research (Section 7.5).

## 7.3 Practical implications

In addition to its analytical and scientific contributions, the thesis has practical implications for procurers introducing electrification to construction procurements. The recommendations draw on the four cases studied and are most relevant where market capacity for emerging technologies is uncertain.

First, instrument choice should match market maturity. Where internal capability is limited and market capacity is uncertain, competitive dialogue can help in absorbing market knowledge before contract terms are finalised and bonus-based instruments can preserve contractor discretion where requirements would be premature. This reduces the risk that requirements become either too loose to drive change or practically unachievable. Once internal knowledge has matured, procurers may be better positioned to move toward direct requirements without the need for dialogue to calibrate feasibility. The procurers in Slakthusområdet and Mäster Samuelsgatan described this scenario while Östra Fyrislund showed that strict requirements can work when capability is sufficient and preparation time is adequate.

Second, the compensation model should be treated as part of the risk-allocation decision, not as a separate procurement choice. Two procurements sharing the same nominal instrument type can lead to different risk distribution depending on how compensation is structured. Slakthusområdet's running-account model with a CO2 equivalent ceiling separated cost risk from performance risk and assigned each to the party better positioned to manage it. Östra Fyrislund's fixed-price model combined with a high penalty placed more cost and compliance exposure on the contractor. The compensation model therefore requires the same attention as the instrument choice itself.

Third, demand signals to the market need to be technically coherent across procurements. Repeated or predictable demand for the same machine categories support contractor investment by extending the time horizon over which costs can be recovered. Switching between categories from one procurement to another can undermine this, even when underlying electrification commitments stays consistent. Procurers planning multiple electrification-oriented procurements should consider how the specifications interact across these projects not only for individual contracts.

Fourth, charging infrastructure and grid access should be treated as procurement design variables, not technical issues to be resolved after tendering. Well mapped grid access assists contractors in planning electrification. Where these aspects are absent or unclear charging could become a source for risk-pricing in tenders. Therefore, procurers should investigate grid capacity, grid connection points and relevant energy-providers before procurement begins in addition to including this information in the tendering material.

Fifth, verification should be designed in parallel with the procurement instrument. Whether a machine is electric is easy to confirm through site inspection. This situation is complicated when there is a mix of electrified and traditional equipment on the site or when there are bonuses/penalties tied to measured electricity usage. Procurers should therefore decide how compliance will be verified before determining the ambition level and precision needed. Otherwise, procurement instruments risk becoming difficult to enforce, and unclear follow-up may increase contractor uncertainty.

Finally, the cases indicate that electrification ambition is more durable if it has institutional backing than when it rests on individual procurer prerogative. Procurements that depend on motivated individuals to introduce electrification-oriented aspects are more fragile than those

based on political mandate or organisational policy. Therefore, consistent demand across multiple procurements is dependent on that the electrification commitment is embedded in the organisation and not only carried by individual procurers.

## 7.4 Limitations and scope of conclusions

One limitation of the study is related to the cases examined. In this thesis four cases were examined, all of which were geographically limited to Sweden. This limits the direct transferability of the findings to procurement contexts where legislation, institutional practice or market conditions differ substantially. Additionally, the comparability between the examined cases is not perfect, they were selected for their variation in procurement instrument application, but they also differ in project type, time and procuring organisation.

Further, the conclusions generalise analytically rather than statistically. The core analysis relates to how a selected group of procurer-side and contractor-side actors interpret and reflect upon relevant aspects of public procurement for electrified construction. Measured outcomes fall outside the scope of the thesis, actual emission reductions and cost effects have not been measured and as such cannot be determined definitively.

Another limitation concerns the absence of relevant actors in the extended value-chain. OEMs, subcontractors and machine owners are not a part of the empirical base since they were not interviewed directly and as such claims about cascading risk through the contracting chain are analytically indicated rather than empirically verified. However, this limitation does not remove the relevance of the thesis's contribution concerning instrument architecture and risk positioning at the procurer–contractor interface, but it does narrow the scope of claims about lower-tier actors.

The rapid pace of technological development in the electric construction equipment market also limits the applicability of the conclusions. Findings about market capacity, feasibility and instrument calibration therefore reflect the situation at the time of the studied cases and may not be directly transferable to a more mature market where similar trade-offs could differ.

Finally, the author's participation in SEEK workshops during the research process shaped familiarity with the broader discourse around electrified procurement. This is acknowledged as a contextual influence on interpretation, particularly regarding themes such as risk sharing, market maturity, charging responsibility and procurement model development.

## 7.5 Further research

The thesis shows several directions where further research would be valuable. Each direction connects either to a limitation of this thesis or to an analytical finding that would benefit from further empirical testing.

First, further research should examine actors in the extended value chain directly. This thesis interviewed procurers and main contractors, but did not directly include subcontractors or

machine owners. The claims about how procurement pathways cascade through the contract chain remain analytically indicated rather than empirically verified. Interview based research on lower-tier actors would improve the clarity on how procurement architecture affects investment, pricing and participation conditions further down the chain, and would test whether the cascading-risk framing developed in this thesis holds in practice.

Second, further research could track the same procuring organisation across multiple procurements to shed light on the transition from competitive dialogue to traditional procurement as organisational competence develops. This thesis observed such a transition through the interviews, but from a single point in time. Following procurers across several procurements would make it possible to examine whether this transition occurs in practice, what triggers it, and how learning is retained within the organisation.

Third, the interpretive analysis presented here would benefit from complementary quantitative research once enough electrified procurements have been conducted. Such a study could include comparative evaluation of emission outcomes, tender participation rates, competition effects, and cost effects across different instrument architectures. This would test whether the risk-positioning logic identified in this thesis translates into observable performance differences.

Finally, the role of institutional anchoring deserves further research. The thesis identified anchoring as a prior condition for electrification-oriented procurement, although this emerged as an exploratory rather than fully developed empirical dimension. A study designed around this particular question could examine how political mandate, organisational policy and individual championship shape the consistency and durability of climate-oriented procurement.

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# Appendix A — Interview Guides

The interview guides were used in Swedish, the language in which all interviews were conducted. They are reproduced here in their original form. The guides shown below are the base versions used across procurer and contractor interviews; minor case-specific probes were added on top of these base guides for each individual interview.

## A.1 Interview Guide — Procurer

Semistrukturerad intervju, ca 45–60 minuter.

**Introduktion och samtycke**

*"Tack för att du tar dig tid att delta. Detta samtal är en del av mitt examensarbete vid KTH inom industriell ekonomi. Jag undersöker hur olika upphandlingsmodeller för elektrifierad bygg- och anläggningsverksamhet, såsom krav, bonus, vite och kombinationer av dessa, påverkar hur risk och ansvar fördelas mellan beställare och entreprenör, och vilka konsekvenser det kan få för deltagande, genomförbarhet och investeringsförutsättningar."*

*"Intervjun är semistrukturerad och beräknas ta ungefär 45 till 60 minuter. Med ditt samtycke spelar jag gärna in samtalet för att underlätta anteckningar och transkribering. Materialet används endast för examensarbetet. I rapporten citeras det ni säger med roll och fall, men ni får möjlighet att läsa igenom citaten innan publicering om ni önskar."*

Checkpunkter: samtycke till inspelning, samtycke till citat med roll, möjlighet att återkomma med följdfrågor.

**1. Roll och involvering**

- Kan du beskriva din roll i projektet och hur du var involverad i upphandlingen?
- Vilka delar av processen var du personligen involverad i: formulera krav, utvärdera anbud, hantera kontraktet, följa upp genomförandet?
- I vilket skede kom du in i processen?

**2. Syfte och bakgrund till elektrifieringskraven**

- Vad var syftet med att inkludera elektrifieringsrelaterade inslag i denna upphandling?
- Vilket problem eller vilken målsättning ville ni adressera?
- Var ambitionen främst miljömässig, strategisk, politisk, marknadsdrivande eller operativ?
- Varför bedömdes just detta projekt vara lämpligt för denna typ av krav?
- Hur resonerade ni kring att gå från generella hållbarhetsambitioner till konkreta krav på faktisk elektrifiering?

**3. Val av upphandlingsutformning**

- Hur resonerade ni när ni valde just denna utformning för att främja elektrifiering?
- Vilka alternativ diskuterades (tydliga kontraktskrav, bonus, vite, utvärderingsincitament eller kombinationer)?
- Vad såg ni som styrkorna med det valda upplägget?
- Fanns det avvägningar eller osäkerheter när upplägget valdes?
- Hur resonerade ni kring kalibreringen av kravnivån — skillnaden mellan vad som är tekniskt möjligt och vad som är kommersiellt genomförbart?

- Om vite användes: vad var den huvudsakliga funktionen — avskräckning, säkerställande av efterlevnad eller signal till marknaden?

**4. Antaganden om marknadens förutsättningar**

- Vilka antaganden hade ni om marknadens möjlighet att möta dessa villkor?
- Hur mogen uppfattade ni att leverantörsmarknaden var?
- Vilka antaganden gjorde ni om tillgång till eldrivna maskiner, laddlösningar, nätanslutning eller arbetssätt?
- Tänkte ni att olika typer av leverantörer kunde påverkas på olika sätt?
- Förde ni någon dialog med marknaden innan upphandlingen utformades?

**5. Deltagande, genomförbarhet och investeringsförutsättningar**

- Hur resonerade ni kring hur utformningen kunde påverka deltagande, genomförbarhet och investeringsförutsättningar för olika leverantörer?
- Tänkte ni att utformningen kunde skapa trösklar för vissa aktörer?
- Reflekterade ni över om mindre företag kunde ha andra förutsättningar än större entreprenörer?
- Hur såg ni på behovet av investeringar i maskiner, laddinfrastruktur eller ny kompetens?
- I vilken utsträckning betraktades det som rimligt att marknadsaktörerna själva skulle bära omställningskostnader?

**6. Kostnader, risker och ansvar**

- Hur tänkte ni kring kostnader, risker och ansvar kopplade till elektrifiering i denna upphandling?
- Vilka typer av risker identifierade ni vid den tidpunkten (kostnad, försening, teknisk prestanda, maskintillgång, laddinfrastruktur, efterlevnad)?
- Vem eller vilka tänkte ni skulle bära de största riskerna i praktiken?
- Reflekterade ni över om risk kunde föras vidare från beställare till huvudentreprenör och vidare till underentreprenörer eller leverantörer?
- Hur tänkte ni kring beställarens egen roll i att skapa praktiska förutsättningar för att kraven skulle kunna uppfyllas (till exempel el eller byggström)?
- Hur resonerade ni kring vad entreprenören kan kontrollera jämfört med vad som ligger utanför deras direkta rådighet?

**7. Uppföljning och verifiering**

- Hur var det tänkt att kraven skulle följas upp, och hur fungerade det i praktiken?
- Vilken typ av dokumentation, rapportering eller verifiering användes?
- Hur resonerade ni kring vad som skulle räknas som tillräcklig bevisning för att kraven uppfylldes?
- Vilken roll spelade vite eller andra uppföljningsmekanismer?
- Fanns det svårigheter i att kontrollera vad som faktiskt levererades?
- Fungerade uppföljningen som ni hade tänkt er?

**8. Konkreta situationer i projektet**

- Kan du ge exempel på konkreta situationer där denna utformning sattes på prov, och hur den fungerade då?
- Fanns det moment där kraven fungerade särskilt väl eller mindre väl?

- Såg ni situationer där utformningen tydligt påverkade entreprenörens planering, val av maskiner eller arbetssätt?
- Uppstod praktiska hinder som påverkade möjligheten att leva upp till ambitionerna?
- Fanns det situationer där uppföljning eller vite aktualiserades på ett tydligt sätt?

**9. Effekter, oavsiktliga konsekvenser och lärdomar**

- När du ser tillbaka, hur effektiv upplever du att utformningen var, och vad skulle du göra annorlunda?
- Fungerade upplägget som ni hade tänkt er?
- Uppstod några oavsiktliga konsekvenser?
- Påverkade utformningen leverantörernas beteende på det sätt ni förväntat er?
- Påverkade den konkurrensen, anbudsnivåerna eller genomförandet på något oväntat sätt?
- Vilka lärdomar tar du med dig till framtida upphandlingar?
- Om ni skulle göra om en liknande upphandling idag, vad skulle ni vilja behålla och vad skulle ni vilja ändra?

**Avslut**

- Finns det något vi inte har pratat om som du tycker är viktigt för min förståelse av upphandlingen?
- Finns det någon kollega eller annan aktör du tycker att jag borde prata med?
- Är det okej om jag återkommer med korta följdfrågor via mejl om något dyker upp i analysen?

## A.2 Interview Guide — Contractor

Semistrukturerad intervju, ca 45–60 minuter.

**Introduktion och samtycke**

*"Tack för att du tar dig tid. Det här samtalet är en del av mitt examensarbete vid KTH inom industriell ekonomi. Jag undersöker hur olika upphandlingsmodeller för elektrifierad anläggning, såsom krav, bonus, vite och kombinationer av dessa, påverkar hur risk och ansvar fördelas mellan beställare och entreprenör, och vad det får för konsekvenser för vilka som kan delta och för hur projekten går att genomföra. Jag är särskilt intresserad av ert perspektiv som entreprenör, både i anbud och i genomförande."*

*"Samtalet är semistrukturerat och tar ungefär 45 till 60 minuter. Jag spelar gärna in och transkriberar om det är okej för dig. Materialet används endast för examensarbetet. I rapporten citeras det ni säger med roll och fall, men ni får möjlighet att läsa igenom citat innan publicering om ni önskar."*

Checkpunkter: samtycke till inspelning, samtycke till namngivet citat eller anonymiserat, möjlighet att återkomma med följdfrågor.

**1. Roll och involvering**

- Kan du beskriva din roll i projektet och när du kom in i processen?
- Vilka delar av processen har du varit involverad i: kvalificering, dialog, anbud, planering, produktion, uppföljning?
- Vilka inom ert företag var med i anbudsarbetet, och hur fördelade ni ansvaret för den elektrifieringsrelaterade delen?

**2. Första uppfattningen av kravställningen**

- När du först läste upphandlingsunderlaget, vad uppfattade du som de mest utmärkande dragen i kravställningen?
- Hur tolkade du balansen mellan ambition och praktisk genomförbarhet på den nivå som efterfrågades?
- Vad var tydligt och vad var otydligt i hur kraven var formulerade?
- Hur uppfattade ni att ambitionsnivån motiverades av beställaren?
- Hur skiljer sig ert arbetssätt i ett kontrakt med skarpa elektrifieringskrav jämfört med en upphandling med delkrav eller bonus?

**3. Anbudsskede, dialog och utvärderingsmodell**

- Hur upplevde ni hela anbudsprocessen jämfört med mer traditionella upphandlingar?
- Om konkurrenspräglad dialog användes: hur påverkade dialogen vad ni såg som rimligt att erbjuda?
- I vilken riktning påverkade processen er slutliga anbudsutformning: pris, scope, riskpositionering eller något annat?
- Fanns det krav som ändrades, mjukades upp eller skärptes som en direkt följd av dialogen?
- Efter anbudet, kände ni att informationsasymmetrin mellan er och beställaren var mindre, oförändrad eller större än i en traditionell upphandling?

**4. Intern bedömning av förutsättningar**

- När ni internt bedömde om kraven var rimliga att uppfylla, vilka frågor var svårast att svara på?
- Hur bedömde ni tillgången på rätt maskiner vid rätt tidpunkt, och vilka beslut tvingade det fram kring inköp, leasing eller uthyrning?
- Behövde ni göra anpassningar i planering, bemanning, logistik eller leverantörskedja för att kunna leverera?
- Hur mycket av den tekniska infrastrukturen (laddning, nätanslutning) hade beställaren förberett, och hur mycket behövde ni lösa själva?
- Fanns det möjliggörande förutsättningar som ni hade förväntat er men inte fick, eller tvärtom?

**5. Deltagande, konkurrens och investering**

- Vilka aktörer tror du realistiskt kunde lägga anbud på den här upphandlingen?
- Hur skiljer sig förutsättningarna för större och mindre aktörer, och för eventuella underentreprenörer, att möta den här typen av krav?
- Vilka investeringar behövde ni göra i maskiner, kompetens eller arbetsmetoder, och var det rimligt i förhållande till projektets storlek och tidshorisont?
- Tror du att den här upphandlingen framför allt gynnar aktörer som redan ligger långt framme, eller öppnar den också en väg för andra att komma ikapp?

**6. Kostnad, risk och ansvar**

- Vilka kostnader och risker bedömde ni som mest betydande när ni räknade på anbudet?
- Hur hanterade ni risker kopplade till maskintillgänglighet, teknisk prestanda, produktivitet, laddning, förseningar och uppföljning?
- Hur uppfattar ni att ansvar fördelades mellan er och beställaren i kontraktet, och stämde den fördelningen när projektet väl startade?
- Fanns det risker som hamnade hos er trots att de var utanför er kontroll? Kan du ge ett exempel?
- Om vite användes: hur påverkade vitesstrukturen er riskbedömning och prissättning?
- Hur stor del av anbudspriset skulle du säga var en riskbuffert för elektrifieringen specifikt?

**7. Nedströmsrisk och underentreprenörer**

- Hur påverkade upphandlingens kravställning era relationer med underentreprenörer, maskinuthyrare och leverantörer?
- Hur implementerar ni elektrifieringskrav i avtal med underentreprenörer?
- Hur hanterar ni en situation där en underentreprenör inte kan leva upp till kravet?
- Har vissa underentreprenörer avstått från att ta uppdrag på grund av kraven?
- Hur tänker ni kring hur mindre aktörer, maskinuthyrare eller enmansföretag ska kunna bära den här typen av investerings- och produktionsrisk?

**8. Uppföljning och verifiering**

- Vilken typ av dokumentation, rapportering och verifiering krävde upphandlingen?
- Hur tydliga var kraven i praktiken, och hur stor blev den administrativa bördan?
- Fokuserade uppföljningen på rätt saker, eller fanns det delar som kändes irrelevanta eller missvisande?
- Hur trovärdig är rapporteringen, och hur skulle den kunna göras mer robust utan oproportionerlig administrativ börda?
- Hur skulle en mindre entreprenör utan stor tjänsteorganisation klara den här rapporteringen?

**9. Konkreta situationer där upphandlingen prövades**

- Kan du beskriva en situation där upphandlingens utformning fungerade särskilt väl?
- Kan du beskriva en situation där den var svår att leva upp till, eller där de praktiska förutsättningarna inte riktigt höll?
- Hur löste ni det när det uppstod en konflikt mellan vad kontraktet krävde och vad som rent produktionstekniskt var möjligt?
- Vilken roll spelade dialogen med beställaren när något oväntat hände?
- I efterhand, vilka delar av upphandlingen hade störst faktisk betydelse för hur projektet gick?

**10. Effekter, oavsiktliga konsekvenser och lärdomar**

- I vilken utsträckning styrde upphandlingens utformning ert beteende på det sätt som beställaren troligen hade tänkt sig?
- Finns det oavsiktliga konsekvenser av hur kraven var formulerade?
- Var upphandlingen tillräckligt realistisk för att fungera i praktiken?
- Om ni fick utforma en liknande upphandling imorgon, hur skulle ni balansera krav, bonus, vite och dialog?
- Vilka lärdomar tar ni med er till framtida upphandlingar av liknande slag?

**Avslut**

- Finns det något vi inte har pratat om som du tycker är viktigt för min förståelse av upphandlingen?
- Finns det någon kollega, underentreprenör eller maskinuthyrare som du tycker att jag borde prata med?
- Är det okej om jag återkommer med korta följdfrågor via mejl om något dyker upp i analysen?